Identification of Beliefs and Preferences from Simple Probability Elicitation

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Abstract

Binned probability elicitation, where an agent allocates probability mass across outcome intervals, is common for eliciting belief distributions in experimental economics, large-scale surveys, and forecasting. I study a simple binned belief elicitation mechanism, the pay-the-probability (PtP) rule, where reports are distorted by risk preferences. Under expected utility with arbitrary concave utility, I derive an inversion formula recovering true beliefs from observed reports, prove that logarithmic utility uniquely implements truthful reporting, and provide a new axiomatic characterization by showing that CRRA utility uniquely exhibits stake invariance. For non-CRRA preferences, stake variation yields a power-scale observational equivalence under common utility with two stakes and nonparametric individual identification under a sieve restriction with three stakes; in both cases, one auxiliary risk-preference measurement resolves the remaining power-scale ambiguity. A penalized minimum-distance estimator exploiting cross-stake consistency and a risk-preference anchor reduces L_1 belief error by 44–70% under the recommended individual regime across Monte Carlo scenarios spanning correct specification to severe misspecification. Ablation confirms both components contribute, out-of-sample validation shows the estimator learns utility structure that generalizes to unseen stakes, and convergence exercises reveal that the individual estimator dominates pooled approaches as a robust default.

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1 Introduction

Binned probability elicitation (Manski, 2004), which involves asking subjects to allocate probability mass across outcome intervals, is a widely-used tool for measuring beliefs. It appears in laboratory experiments on strategic thinking and belief formation, in field experiments on technology adoption, and in information provision experiments that stand alone or are embedded into large-scale surveys. Further, it appears in major household surveys such as the Federal Reserve Bank of New York's Survey of Consumer Expectations (SCE), the Health and Retirement Study (HRS), and the Survey of Economic Expectations (SEE). In experimental settings the mechanism is routinely incentivized and Drobot et al. (2026a) demonstrate that replacing flat participation fees with marginal incentives for accuracy in an SCE-style environment materially alters reported inflation expectations. Drobot et al. (2026b) extend this finding to a systematic comparison of incentive designs, showing that simple, non-proper scoring rules like PtP can outperform theoretically incentive-compatible, but cognitively taxing mechanisms. These findings raise a natural identification question: if belief elicitation is incentivized via the pay-the-probability (PtP) rule, what can we recover from the resulting reports? Under essentially all levels of risk aversion for a broad class of utility functions, reported probabilities diverge systematically from the agent's true belief. This introduces a structured distortion shaped by the interaction of risk preferences with the mechanism's payoff structure (Manski, 2004; Winkler and Murphy, 1970; Kadane and Winkler, 1988; Harrison et al., 2017).

I show that this distortion is a source of identifying power. Under expected utility with arbitrary concave utility, the optimal distributional forecast under PtP exists, is unique, and admits an explicit inversion formula recovering true beliefs from stated beliefs, conditional on knowing the agent's utility. For non-CRRA preferences, varying the stake size generates observable changes in reports that trace out the shape of marginal utility, enabling its nonparametric recovery. The stake plays the role of an exclusion restriction: it shifts the mapping from true to stated beliefs without affecting beliefs themselves. The methodological parallel is the structural auction literature of Guerre et al. (2000), where bid variation identifies private values; here, stake variation identifies utility, and inversion recovers beliefs. Guerre et al. (2009) extended the auction framework to identify risk aversion under exclusion restrictions (variables that shift the bid distribution without affecting valuations); our stake variation serves exactly this purpose. The connection to the broader nonparametric identification literature surveyed by Matzkin (2007) and Athey and Haile (2007) is direct: my identification argument exploits the functional structure of the optimality conditions to recover latent parameters from observable behavior.

I establish five main results. First, Theorem 1 characterizes optimal reporting under PtP for arbitrary EU preferences: the inversion formula

$$p_k = \frac{1/u'(c\, g_k^*)}{\sum_{j \in S(\boldsymbol{p})} 1/u'(c\, g_j^*)}, \qquad k \in S(\boldsymbol{p}),$$

generalizes the CRRA-specific formula explored in Drobot et al. (2026b) to arbitrary concave utility. Harrison et al. (2017) study the analogous question for the quadratic scoring rule, showing that under CRRA preferences the distortion from risk aversion is small enough to be safely ignored for empirically plausible parameter values. The PtP case is fundamentally different: the distortion *can be* large, but it is structured and invertible, and stake variation turns it into a source of nonparametric identification. Second, Theorem 2 proves that logarithmic utility is the unique preference under which PtP implements truthful reporting—the first such characterization within the full EU class. Third, Theorem 7 shows that stake invariance characterizes the CRRA family, providing the theoretical foundation for the empirical sensitivity documented by Armantier and Treich (2013). Fourth, Theorem 8 establishes the population-level power-scale observational equivalence under two stakes and clarifies the role of an auxiliary risk measurement. Fifth, and most important, Theorem 9 shows that with $M \geq 3$ stakes and a sieve smoothness restriction ($d \leq (M-1)(K-1)$, where K is the number of bins), we can recover an agent's utility and beliefs from their own reports with no cross-agent homogeneity required. For three stakes and ten bins, $d \leq 18$, accommodating essentially any smooth utility. To my knowledge, this is the first result in the belief elicitation literature achieving joint nonparametric identification of beliefs and preferences at the individual level.

A penalized minimum-distance (PMD) estimator makes the identification strategy operational. Under the recommended individual regime, it reduces L_1 belief error by 44–70% across Monte Carlo scenarios spanning correct specification to severe misspecification, with 85–97% of agents individually improving. Out-of-sample validation—holding out a stake and predicting beliefs at the withheld stake—confirms the estimator learns genuine utility structure that transfers within the training range. The individual estimator matches or outperforms correctly pooled estimation even when pooling is justified, making it a dominant strategy: the researcher need not test or assume utility homogeneity.

The dominant approaches to resolving this identification problem follow one of two paths, neither fully satisfactory. The first assumes risk neutrality, under which strictly proper scoring rules (the quadratic scoring rule (Brier, 1950) and the logarithmic scoring rule (Savage, 1971) being the canonical examples) induce truthful reporting as a dominant strategy. Gneiting and Raftery (2007) provide a comprehensive characterization of the class of strictly proper scoring rules. This approach is theoretically clean but empirically fragile. A large body of evidence, beginning with Winkler and Murphy (1970), Matheson and Winkler (1976) and Kadane and Winkler (1988) and synthesized in the surveys of Schotter and Trevino (2014) and Schlag et al. (2015), documents that risk aversion systematically distorts reports under proper scoring rules. Peysakhovich and Plagborg-Møller (2012) characterize the direction of distortion for non-binary outcomes: reported probability ratios are "shaded toward 1," a compression effect on the simplex. Harrison et al. (2017) study distortions in continuous outcome spaces and argue that, under empirically plausible levels of risk aversion, the bias can be substantial for tail probabilities.

The second approach, *binarization*, converts scoring rule payoffs into probabilities of receiving a fixed prize, thereby theoretically eliminating the influence of risk preferences (Smith, 1961; Allen, 1987; Hossain and Okui, 2013). The idea, which traces to Savage (1971) and was formalized by Allen (1987) and McKelvey and Page (1990), is elegant: by mapping monetary payoffs into lottery probabilities, the agent faces a risk-free (in the sense of expected utility) decision problem, and truthful reporting is restored. Yet binarization introduces significant cognitive complexity, requiring agents to understand both a nonlinear scoring function and a lottery mechanism. Danz et al. (2022) provide a striking demonstration that the binarized scoring rule (BSR) fails *behavioral* incentive compatibility: providing subjects with information about BSR incentives *increases* deviation from truthful reporting, and only 15% of subjects consistently report truthfully. Trautmann and van de Kuilen (2015) reach a similar conclusion in their experimental comparison of elicitation methods, finding that simpler mechanisms often perform as well as or better than theoretically superior ones.

A third strand of the literature has sought to characterize optimal reporting under general preferences without restricting attention to proper scoring rules. Offerman et al. (2009) propose a "truth serum" correction that adjusts proper scoring rule reports for estimated risk attitudes, requiring a separate risk elicitation task. Karni (2009) develops an incentive-compatible mechanism based on a Becker–DeGroot–Marschak (BDM) procedure that is theoretically robust to risk preferences but operationally complex. Chambers and Lambert (2021) develop a dynamic belief elicitation protocol capable of recovering entire belief structures using menus of decision problems. Lambert (2011) characterizes incentive-compatible compensation structures for agents with arbitrary risk preferences, showing that these decompose into families of proper scoring rules.

The closest antecedent is Chambers et al. (2019), who study the same broad problem of recovering beliefs in the presence of risk-preference distortions. They characterize optimal reports under continuous, strictly-proper scoring rules for agents with general preferences, and recover true beliefs for agents with CARA and CRRA utilities from distorted reports conditional on knowing the curvature parameter. Their joint elicitation result is a direct mechanism-design solution: the agent reports both her belief and risk-aversion parameter, and randomization over two quadratic scoring rules makes truthful reporting a strict best response. My work differs in both approach and identification target. First, the identification here is econometric rather than direct-mechanism based: a single, cognitively simple PtP rule is used at multiple stakes, and stake variation identifies utility shape from observed reporting behavior. I then use one auxiliary risk-preference measurement (e.g., from a standard incentivized task) to pin the remaining power-scale ambiguity. This differs from requiring an in-mechanism self-report of the preference parameter or randomization across scoring rules. Second, their inversion results rely on a known parametric preference class (CARA or CRRA), whereas Theorem 9 delivers individual-level nonparametric identification of utility and beliefs under a sieve smoothness restriction. Substantively, this accommodates smooth concave utilities with nonconstant curvature, for example HARA-type, expo-power, or mixtures of power terms (including the mixed-CRRA forms used in the simulations). Third, Chambers et al. (2019) work within the proper-scoring-rule paradigm, treating risk preferences as a nuisance to be corrected, while this paper works outside that paradigm and treats the non-properness of PtP as the source of identifying variation. The tradeoff is that the present analysis is currently confined to expected utility, whereas their duality framework extends to ambiguity aversion and other non-expected-utility preferences; Section 6 discusses first steps toward relaxing this restriction.

Section 2 introduces the model. Section 3 develops the characterization, uniqueness, and stake-invariance results. Section 4 presents the identification theory and estimation. Section 5 provides Monte Carlo evidence. Section 6 addresses robustness and extensions. Section 7 concludes. The appendix contains additional simulation exercises.====================

2 Model

2.1 Environment

A random variable π takes values in $\mathbb{R}$. The outcome space is partitioned into $K \geq 3$ disjoint bins,

$$\{b_k\}_{k=1}^{K}, \qquad \bigcup_{k=1}^{K} b_k = \mathbb{R}, \quad b_k \cap b_j = \emptyset \text{ for } k \neq j.$$

An agent holds subjective beliefs represented by a probability vector

$$\boldsymbol{p} = (p_1, \ldots, p_K) \in \Delta^{K-1},$$

where $p_k \geq 0$ for all k and $\sum_{k=1}^{K} p_k = 1$. Let $S(\boldsymbol{p}) := \{k : p_k > 0\}$ denote the *support* of beliefs, and assume $|S(\boldsymbol{p})| \geq 2$ (the agent regards at least two bins as possible).

The bins $\{b_k\}$ function as discrete states: the agent's payoff depends only on which bin the realization falls in. For concreteness, the SCE uses $K = 10$ bins covering inflation outcomes at 1-percentage-point intervals; the HRS uses similar binned formats for survival and income expectations. The identification results require only the bin probabilities, not the bin endpoints, so the bins need not be bounded.

2.2 Pay-the-Probability Rule

The agent reports a probability vector $\boldsymbol{g} = (g_1, \ldots, g_K) \in \Delta^{K-1}$. If the realized outcome π falls in bin b_k, the agent receives a monetary payment

$$S(\boldsymbol{g}, \pi) = c \cdot g_k \quad \text{if } \pi \in b_k, \tag{1}$$

where $c > 0$ is the *stake size* (a known, fixed parameter of the mechanism).

Under risk neutrality, the agent's expected payoff is $\sum_k p_k \cdot c \cdot g_k = c \cdot \boldsymbol{p}'\boldsymbol{g}$, which is linear in $\boldsymbol{g}$ and maximized by placing all mass on the modal bin. The rule is therefore *not* incentive compatible under risk neutrality. This is the basis for its standard dismissal in the mechanism design literature. The contribution of this paper is to show that this apparent defect is, under risk aversion, a source of identifying power.

2.3 Preferences

The agent has expected utility preferences with a Bernoulli utility function u over monetary payoffs. I maintain the following assumptions throughout:

Assumption 1 (Utility regularity)**.** *The function $u : (0, \infty) \to \mathbb{R}$ satisfies:*

(i) u is twice continuously differentiable on $(0, \infty)$;

(ii) $u'(x) > 0$ for all $x > 0$ (strict monotonicity);

(iii) $u''(x) < 0$ for all $x > 0$ (strict concavity);

(iv) $\lim_{x \to 0^+} u'(x) = +\infty$ and $\lim_{x \to \infty} u'(x) = 0$ (Inada conditions).

The Inada conditions ensure that any bin with positive probability receives positive reported mass: the upper condition ($u'(0^+) = +\infty$) prevents corner solutions on the support of beliefs, while the lower condition ($u'(\infty) = 0$) ensures well-behaved asymptotics. I normalize baseline wealth to zero for expositional clarity; all results extend to $u(w + c\, g_k)$ with $w > 0$, as discussed in Section 6.

Under these preferences, the agent's problem is

$$\max_{\boldsymbol{g} \in \Delta^{K-1}} \; V(\boldsymbol{g}) := \sum_{k=1}^{K} p_k \, u(c\, g_k). \tag{2}$$

3 Optimal Reporting, Uniqueness, and Stake Invariance

3.1 Existence, Uniqueness, and Characterization

Theorem 1 (General EU Characterization)**.** *Let u satisfy Assumption 1, let $c > 0$, and let $\boldsymbol{p} \in \Delta^{K-1}$ with $|S(\boldsymbol{p})| \geq 2$. Then:*

(i) ***Existence, uniqueness, and support recovery:*** *The problem (2) has a unique solution $\boldsymbol{g}^*$. The optimal report satisfies*

$$g_k^* = 0 \quad \Longleftrightarrow \quad p_k = 0. \tag{3}$$

In particular, restricted to the support $S(\boldsymbol{p})$, the solution is interior: $g_k^* > 0$ for all $k \in S(\boldsymbol{p})$.

(ii) ***First-order characterization:*** *For all $k \in S(\boldsymbol{p})$, the optimal report satisfies*

$$p_k \, u'(c\, g_k^*) = \lambda, \qquad k \in S(\boldsymbol{p}), \tag{4}$$

where $\lambda > 0$ is the Lagrange multiplier on the simplex constraint.

(iii) ***Monotonicity:*** *For all $k, j \in S(\boldsymbol{p})$,*

$$g_k^* > g_j^* \quad \Longleftrightarrow \quad p_k > p_j.$$

(iv) ***Inversion formula:*** *True beliefs are recovered from optimal reports by*

$$p_k = \frac{1/u'(c\, g_k^*)}{\displaystyle\sum_{j \in S(\boldsymbol{p})} 1/u'(c\, g_j^*)}, \quad k \in S(\boldsymbol{p}); \qquad p_k = 0, \quad k \notin S(\boldsymbol{p}). \tag{5}$$

Proof. **Part (i).** If $p_k = 0$, allocating mass to bin k wastes resources; hence $g_k^* = 0$. Conversely, for $k \in S(\boldsymbol{p})$, the Inada condition $u'(0^+) = +\infty$ ensures that shifting ε mass to a bin with $g_k = 0$ yields infinite marginal gain, so $g_k^* > 0$. The support face is compact, V is continuous, and strict concavity of u yields uniqueness. (Full details in Section A.1.)

Part (ii). On the support, the KKT conditions with simplex multiplier give $p_k\, u'(c\, g_k^*) \cdot c = \mu$ for all $k \in S(\boldsymbol{p})$; writing $\lambda := \mu/c$ yields (4).

Part (iii). From (4), $p_k > p_j$ implies $u'(c\, g_k^*) < u'(c\, g_j^*)$, and strict concavity gives $g_k^* > g_j^*$.

Part (iv). From (4): $p_k = \lambda/u'(c\, g_k^*)$. Summing and inverting gives (5). $\square$

Remark 1 (Interpretation)**.** *The inversion formula* (5) *states that true beliefs are proportional to* $1/u'$ *evaluated at the reported allocation. The formula requires knowledge of* u*. I discuss how to recover* u *in 4. Intuitively, this result tells us that we can always map from stated beliefs to the true underlying belief distribution if we can first identify respondents' preferences.*

Remark 2 (Reported zeros and support recovery)**.** *The support recovery property* (3) *has a sharp implication: under Inada conditions, observing* $g_k^* = 0$ *in the data implies* $p_k = 0$*. That is, the agent assigns zero probability to bin* k*. In applications, however, reported zeros may reflect rounding or coarse reporting rather than dogmatic beliefs (Manski and Molinari, 2010). Two approaches address this mismatch without altering the model structure. First, one may treat observed reports as coarsened versions of a latent interior optimum and apply* ε*-smoothing: replace* $\tilde{g}_k$ *with* $\max\{\tilde{g}_k, \varepsilon\}/\sum_j \max\{\tilde{g}_j, \varepsilon\}$ *before inversion, with* ε *small. Second, if the boundary marginal utility is finite (i.e., Inada is relaxed), reported zeros imply one-sided bounds rather than point identification: using pay-the-probability,* $g_k^* = 0$ *implies* $p_k \leq \lambda/u'(0^+)$*; more generally, under payoff map* s*,* $p_k \leq \lambda/[u'(c\, s(0^+))\, s'(0^+)]$*.*

3.2 Truthful Reporting and the Uniqueness of Logarithmic Utility

I now ask: for which utility functions does the pay-the-probability rule implement truthful reporting?

Theorem 2 (Log Utility Uniqueness)**.** *Let* $K \geq 3$ *and let* u *satisfy Assumption 1. The optimal report satisfies* $\boldsymbol{g}^* = \boldsymbol{p}$ *for all* $\boldsymbol{p} \in \Delta^{K-1}$ *with* $|S(\boldsymbol{p})| \geq 2$ *and all* $c > 0$ *if and only if* u *is logarithmic:*

$$u(x) = A \ln x + B, \qquad A > 0,\ B \in \mathbb{R}.$$

Proof. **Sufficiency.** Suppose $u(x) = A \ln x + B$. Then $u'(x) = A/x$. The first-order condition (4) on the support requires

$$p_k u'(c g_k^*) = \lambda \quad \text{for all } k \in S(\boldsymbol{p}).$$

Substituting $u'(x) = A/x$ gives

$$p_k \frac{A}{c g_k^*} = \lambda,$$

so

$$g_k^* = \frac{A p_k}{c\lambda}.$$

Summing over k and using $\sum_k g_k^* = 1$ yields

$$1 = \sum_k g_k^* = \frac{A}{c\lambda} \sum_k p_k = \frac{A}{c\lambda},$$

since $\sum_k p_k = 1$. Hence $c\lambda = A$, and therefore

$$g_k^* = p_k.$$

Necessity. Suppose $\boldsymbol{g}^* = \boldsymbol{p}$ for all $\boldsymbol{p} \in \Delta^{K-1}$ with $|S(\boldsymbol{p})| \geq 2$ and all $c > 0$. In particular, this holds for every interior $\boldsymbol{p} \in \operatorname{int}(\Delta^{K-1})$.

Fix $c > 0$. The first-order condition with $g_k^* = p_k$ becomes

$$p_k \, u'(c p_k) = \lambda(\boldsymbol{p}, c) \quad \text{for all } k.$$

Since the multiplier $\lambda(\boldsymbol{p}, c)$ is the same across coordinates, we obtain

$$p_j u'(c p_j) = p_k u'(c p_k) \quad \text{for all } j, k.$$

Because $K \geq 3$, for any $s, t \in (0, 1)$ with $s + t < 1$, the vector

$$\boldsymbol{p} = \left(s, \ t, \ \frac{1 - s - t}{K - 2}, \ \ldots, \ \frac{1 - s - t}{K - 2}\right) \in \operatorname{int}(\Delta^{K-1})$$

is interior (all components strictly positive). Applying the equality above to this vector yields

$$s \, u'(cs) = t \, u'(ct).$$

Since this holds for arbitrary $s, t \in (0, 1)$, the function

$$\varphi_c(x) := x \, u'(cx)$$

must be constant on $(0, 1)$. Hence there exists a scalar $\alpha(c)$ such that

$$x \, u'(cx) = \alpha(c) \quad \text{for all } x \in (0, 1).$$

Solving for $u'(cx)$ gives

$$u'(cx) = \frac{\alpha(c)}{x}.$$

Let $z = cx$, so $x = z/c$. Then for $z \in (0, c)$,

$$u'(z) = \frac{\alpha(c)}{z/c} = \frac{\alpha(c)c}{z}.$$

Thus, for each fixed $c > 0$, we have

$$u'(z) = \frac{\alpha(c)c}{z} \quad \text{for } z \in (0, c).$$

Now let $c_1, c_2 > 0$. On the overlapping interval $(0, \min\{c_1, c_2\})$, both representations must agree:

$$\frac{\alpha(c_1)c_1}{z} = \frac{\alpha(c_2)c_2}{z}.$$

Hence

$$\alpha(c_1)c_1 = \alpha(c_2)c_2.$$

Therefore the product $\alpha(c)c$ is independent of c. Denote this constant by $A > 0$. We conclude

$$u'(z) = \frac{A}{z} \quad \text{for all } z > 0.$$

Integrating yields

$$u(z) = A \ln z + B.$$

□

Remark 3 (Binary elicitation). *The requirement $K \geq 3$ is essential. When $K = 2$, the truthful reporting condition reduces to a functional equation admitting a family of solutions parameterized by an arbitrary symmetric function, so log utility is not uniquely selected. Binary elicitation does not generate sufficient variation to identify utility curvature. Details are in Section A.4.*

3.3 The CRRA Special Case

When u belongs to the CRRA family, the general characterization yields closed-form expressions.

Corollary 3 (CRRA Distortion). *Let $u(x) = x^{1-\rho}/(1-\rho)$ for $\rho > 0$, $\rho \neq 1$ (or $u(x) = \ln x$*

for $\rho = 1$). The unique optimal report is

$$g_k^*(\rho) = \frac{p_k^{1/\rho}}{\sum_{j=1}^K p_j^{1/\rho}}, \qquad k = 1, \dots, K. \tag{6}$$

As $\rho \to 0^+$ (risk neutrality), $\boldsymbol{g}^*$ concentrates on the mode; at $\rho = 1$, $g_k^* = p_k$; as $\rho \to \infty$, $g_k^* \to 1/K$. The inverse mapping is

$$p_k = \frac{(g_k^*)^\rho}{\sum_{j=1}^K (g_j^*)^\rho}. \tag{7}$$

Proof. For CRRA, $u'(x) = x^{-\rho}$. From (4): $p_k(c\, g_k^*)^{-\rho} = \lambda$, so $g_k^* = c^{-1}(p_k/\lambda)^{1/\rho}$. The simplex constraint yields (6); c cancels. $\square$

Equation (6) reveals that CRRA preferences induce a power transformation of beliefs. The mapping

$$p_k \;\mapsto\; \frac{p_k^{1/\rho}}{\sum_j p_j^{1/\rho}}$$

is known in the information-theory literature as an escort transformation. Log utility ($\rho = 1$) is uniquely characterized by invariance under this transformation, yielding proportional truthfulness $g_k^* = p_k$. For $\rho > 1$ (greater risk aversion), the transformation flattens beliefs toward the uniform distribution, while for $\rho < 1$ it concentrates mass toward higher-probability states. Thus, CRRA curvature induces a systematic and monotone distortion of beliefs, with logarithmic utility emerging as the unique fixed point of the power family.

3.4 Invertibility of the Report Mapping

Lemma 4 (Local invertibility)**.** *For any $\boldsymbol{p} \in \operatorname{int}(\Delta^{K-1})$ and $c > 0$, $\boldsymbol{p} \mapsto \boldsymbol{g}^*(\boldsymbol{p}, c)$ is C^1 with full-rank Jacobian on the simplex tangent space.*

The proof is a standard implicit function theorem argument applied to the FOC system; it appears in Section A. The next result strengthens local invertibility to global injectivity.

Proposition 5 (Global injectivity and inversion)**.** *Fix $c > 0$. On $\operatorname{int}(\Delta^{K-1})$, $\boldsymbol{p} \mapsto \boldsymbol{g}^*(\boldsymbol{p}, c)$ is globally injective, with inverse*

$$p_k = \frac{1/u'(c\, g_k)}{\sum_{j=1}^K 1/u'(c\, g_j)}, \qquad k = 1, \dots, K. \tag{8}$$

Proof. If $\boldsymbol{g}^*(\boldsymbol{p}, c) = \boldsymbol{g}^*(, c) =: \boldsymbol{g}$, the FOCs give $p_k/q_k = \lambda_p/\lambda_q$ for all k, so $\boldsymbol{p} = \alpha$; summing forces $\alpha = 1$. Inversion follows from $p_k = \lambda/u'(c\, g_k)$ and normalization. □

3.5 Stake Invariance and the Characterization of CRRA

Corollary 3 reveals that under CRRA, the optimal report does not depend on c. I now show this property *characterizes* the CRRA family.

Definition 6 (Stake invariance)**.** *The pay-the-probability rule exhibits* stake invariance *under utility u if, for every $\boldsymbol{p} \in \Delta^{K-1}$ with $|S(\boldsymbol{p})| \geq 2$ and every $c_1, c_2 > 0$, the optimal reports coincide: $\boldsymbol{g}^*(c_1) = \boldsymbol{g}^*(c_2)$.*

Theorem 7 (Stake Invariance Characterizes CRRA)**.** *Let $K \geq 3$ and let u satisfy Assumption 1. Stake invariance holds if and only if u belongs to the CRRA family:*

$$u(x) = \begin{cases} \dfrac{x^{1-\rho}}{1-\rho} + B, & \rho > 0,\ \rho \neq 1, \\ A \ln x + B, & \rho = 1, \end{cases}$$

for constants $A > 0$, $B \in \mathbb{R}$, $\rho > 0$.

Proof. **Sufficiency.** If u is CRRA, $u'(x) = Ax^{-\rho}$, and the report (6) is independent of c.

Necessity. Assume stake invariance: for every $\boldsymbol{p} \in \operatorname{int}(\Delta^{K-1})$ and $c_1, c_2 > 0$, $\boldsymbol{g}^*(\boldsymbol{p}, c_1) = \boldsymbol{g}^*(\boldsymbol{p}, c_2)$.

Fix any $c_1, c_2 > 0$. Take arbitrary $x, y \in (0, 1)$ with $x + y < 1$, and choose any interior report vector $\boldsymbol{g}$ with $g_1 = x$, $g_2 = y$ (remaining coordinates positive, summing to $1 - x - y$). Define beliefs by the inversion formula at stake c_1:

$$p_k := \frac{1/u'(c_1 g_k)}{\sum_j 1/u'(c_1 g_j)}.$$

By Theorem 1, this $\boldsymbol{g}$ is the unique optimum under $(\boldsymbol{p}, c_1)$. By stake invariance, the same $\boldsymbol{g}$ is also optimal under $(\boldsymbol{p}, c_2)$. Applying FOC ratios for bins 1 and 2:

$$\frac{u'(c_1 x)}{u'(c_1 y)} = \frac{p_2}{p_1} = \frac{u'(c_2 x)}{u'(c_2 y)}.$$

Hence, for all $x, y \in (0, 1)$ with $x + y < 1$, the ratio $u'(cx)/u'(cy)$ is independent of c; by continuity, this extends to all $x, y \in (0, 1)$.

Set $f := u'$. Taking logs and differentiating with respect to c:

$$x\,\frac{f'(cx)}{f(cx)} = y\,\frac{f'(cy)}{f(cy)} \quad \forall\, c > 0,\ \forall\, x, y \in (0,1).$$

So for each fixed c, the quantity $x\,f'(cx)/f(cx)$ is constant in x; write $x\,f'(cx)/f(cx) = A(c)$ for $x \in (0,1)$. Substituting $t = cx \in (0,c)$: $t\,f'(t)/f(t) = c\,A(c) =: B(c)$ for $t \in (0,c)$. For $c_1 < c_2$, this identity holds on $(0,c_1)$ for both c_1, c_2, so $B(c_1) = B(c_2)$. Therefore $B(c) \equiv \alpha$ is constant, and $f'(t)/f(t) = \alpha/t$ for all $t > 0$. Integrating: $f(t) = A\,t^{\alpha}$ with $A > 0$. Since u is strictly concave, $f = u'$ is strictly decreasing, hence $\alpha < 0$. Setting $\rho := -\alpha > 0$ gives $u'(t) = A\,t^{-\rho}$, i.e., the CRRA family. □

Remark 4 (Testable implication)**.** *Theorem 7 provides an immediate empirical test: if reports do not change across stakes, CRRA is supported. If they do change, CRRA is rejected and the pattern of change identifies the curvature of u'.*

4 Nonparametric Identification of Beliefs and Preferences

I now develop the main identification results. For non-CRRA utility, stake variation generates observable changes in reports that trace out the shape of u', enabling its nonparametric recovery.

4.1 Setup and Notation

Consider a population of agents indexed by $i \in \mathcal{I}$. Each agent holds beliefs $\boldsymbol{p}^i \in \Delta^{K-1}$ with $|S(\boldsymbol{p}^i)| \geq 2$. The mechanism is administered under $M \geq 2$ stake sizes $c_1, \ldots, c_M$. Under stake c_s, agent i reports $\boldsymbol{g}^{i,s}$. By Theorem 1, $g_k^{i,s} > 0$ if and only if $p_k^i > 0$. Identification operates on the support of each agent's beliefs.

Assumption 2 (Non-CRRA)**.** *The utility function u_i is not CRRA. Equivalently (by Theorem 7), optimal reports differ across stakes for generic beliefs.*

4.2 Population Two-Stake Power-Scale Ambiguity

I first isolate a population-level observational equivalence class under the assumption that all agents share a common utility function. This is the source of the power-scale ambiguity that later motivates the auxiliary risk-preference anchor.

Assumption 3 (Common utility)**.** *All agents share the same utility function u satisfying Assumption 1.*

Assumption 4 (Belief heterogeneity and two-point exact-overlap connectivity). *For each agent-stake pair (i, s), define the finite evaluation-point set*

$$X_{i,s} := \{c_s\, g_k^{i,s} : k \in S(\boldsymbol{p}^i)\} \subset (0, \infty).$$

Let $\mathcal{X} := \bigcup_{i,s} X_{i,s}$. Assume:

(i) $\mathcal{X}$ is dense in a compact interval $[a, b]$ with $0 < a < b < \infty$.

(ii) The graph $\mathcal{G}$ with vertices (i, s) and edges

$$(i, s) \sim (i', s') \quad \Longleftrightarrow \quad |X_{i,s} \cap X_{i',s'}| \geq 2$$

is connected.

The two-stake first-order conditions imply within-stake belief ratios $p_k^i/p_j^i = u'(c_s g_j^{i,s})/u'(c_s g_k^{i,s})$. Since beliefs do not change across stakes,

$$\frac{u'(c_1\, g_j^{i,1})}{u'(c_1\, g_k^{i,1})} = \frac{u'(c_2\, g_j^{i,2})}{u'(c_2\, g_k^{i,2})}, \qquad \text{for all } k, j, i, \tag{9}$$

a system of functional equations in the unknown u' evaluated at observable points, with beliefs eliminated entirely. These restrictions immediately imply the power-scale observational equivalence in Theorem 8. Under additional overlap-normalization conditions (Assumption 6 in Appendix A.3), the appendix proves a full converse characterization of the observational-equivalence class. .

Theorem 8 (Population-Level Power-Scale Observational Equivalence under Two Stakes). *Suppose Assumptions 1 to 4 hold and agents are observed under two distinct stakes $c_1 \neq c_2$. Then:*

(i) If $(u, \{\boldsymbol{p}^i\})$ rationalizes the observed reports, then for any constants $\alpha > 0$ and $\beta > 0$, the transformed pair defined by

$$\tilde{u}'(x) = \beta\, [u'(x)]^{\alpha}, \qquad \tilde{p}_k^i = \frac{(p_k^i)^{\alpha}}{\sum_j (p_j^i)^{\alpha}}$$

also rationalizes the same reports. Consequently, two-stake cross-stake variation alone does not point identify u' or beliefs, even under common utility.

(ii) One auxiliary risk-preference measurement pins α within this power-scale equivalence class. For instance, an observed value of $\mathrm{RRA}(x_0) = -x_0\, u''(x_0)/u'(x_0)$ *at any* $x_0 \in$

$[a,b]$ determines α, since $\widetilde{\mathrm{RRA}}(x_0) = \alpha\,\mathrm{RRA}(x_0)$. The multiplicative constant β cancels in the inversion formula (5), so the auxiliary measurement selects a unique representative of the power-scale family up to the innocuous constant β.

Part (i) gives the constructive power-scale equivalence. The converse direction is established in Appendix A.3 (Proposition 11) under Assumption 6.

Proof. **Part (i).** Fix any $\alpha > 0$ and $\beta > 0$, and define

$$\tilde{u}'(x) := \beta\,[u'(x)]^{\alpha}, \qquad \tilde{p}^i_k := \frac{(p^i_k)^{\alpha}}{\sum_j (p^i_j)^{\alpha}}.$$

If the original model rationalizes reports $\{\boldsymbol{g}^{i,s}\}$, then for every i, s, k on the support,

$$p^i_k\, u'(x^{i,s}_k) = \lambda^{i,s}, \qquad x^{i,s}_k := c_s g^{i,s}_k.$$

Hence

$$\tilde{p}^i_k\, \tilde{u}'(x^{i,s}_k) = \frac{\beta}{Z_i}\,(p^i_k)^{\alpha}[u'(x^{i,s}_k)]^{\alpha} = \frac{\beta}{Z_i}\,(\lambda^{i,s})^{\alpha},$$

where $Z_i := \sum_j (p^i_j)^{\alpha}$. The right-hand side is independent of k, so the same reports satisfy the tilded FOCs.

Part (ii). Let $\psi := \ln \circ u'$. Under the power-scale family from part (i),

$$\tilde{\psi}(x) := \ln \tilde{u}'(x) = \ln\beta + \alpha\,\psi(x).$$

Hence $\widetilde{\mathrm{RRA}}(x_0) = -x_0\,\tilde{\psi}'(x_0) = \alpha\,[-x_0\,\psi'(x_0)] = \alpha\,\mathrm{RRA}(x_0)$, so the auxiliary observation determines α within the equivalence class. With α pinned, the remaining multiplicative constant β cancels in (5).

Assumption 4 is combined with Assumption 6 in Appendix A.3 (Proposition 11), which proves the full converse characterization.

□

Remark 5 (Primitive conditions for Assumption 4)**.** *Assumption 4 is a population-level richness condition. The two-point exact-overlap requirement in part (ii) is stronger than interval/range overlap and is stated for a clean converse characterization in the continuum-population idealization. In finite samples, exact equality of evaluation points is not expected; the PMD estimator replaces this with approximate cross-stake consistency and smoothing/aggregation across agents.*

Remark 6 (The CRRA boundary)**.** *When u is CRRA, reports are stake-invariant and stake variation provides no identifying power. Under sufficient independent variation in beliefs (e.g., a population with heterogeneous priors whose belief distribution is known or independently estimated) or auxiliary risk preference tasks, the CRRA parameter ρ can be identified from the pattern of reports (the degree of compression toward uniformity). Absent such external variation, CRRA curvature and beliefs are not separately identified from reports alone.*

4.3 Individual-Level Identification with Three or More Stakes

The common utility assumption (Assumption 3) may be undesirable when agents plausibly differ in risk preferences. I now show it can be dispensed with under three or more stakes, provided a mild smoothness restriction holds. The identification challenge is that each agent's reports provide only finitely many evaluation points for $\psi_i := \ln \circ u_i'$, and without further restriction, a continuum of strictly decreasing functions could interpolate through any finite set of points. I resolve this using a sieve restriction.

Assumption 5 (Smoothness of log-marginal utility)**.** *$\psi_i := \ln \circ u_i'$ belongs to a known d-dimensional linear function space Ψ_d: $\psi_i(x) = \sum_{l=1}^{d} \theta_l^i \phi_l(x)$ for known basis functions $\phi_1, \dots, \phi_d : (0, \infty) \to \mathbb{R}$ and parameter vector $\theta^i \in \Theta \subset \mathbb{R}^d$, where Θ is the set ensuring ψ_i is strictly decreasing.*

Remark 7 (Sieve as nonparametric approximation)**.** *Assumption 5 is a sieve restriction in the sense of Newey (1997), not a parametric assumption. The dimension d is bounded by the survey or experimental design ($d \leq (M-1)(K-1)$) and grows as the number of stakes M or bins K increases. As M or K grow, the admissible sieve dimension grows, and the sieve approximates any smooth, strictly decreasing function, so the restriction becomes vacuous in the limit (Chen, 2007). For $M = 3$ and $K = 10$, $d \leq 18$: far more flexible than CRRA or CARA (each of which has one free shape parameter under the convention that $\{\phi_l\}$ excludes constants). In practice, choosing $d < (M-1)(K-1)$ provides overidentifying restrictions that enable specification testing and improve finite-sample robustness. The sieve dimension can be selected by standard model selection criteria or by overidentification testing.*

Theorem 9 (Individual-Level Identification under Sieve Smoothness)**.** *Let $K \geq 3$, $M \geq 3$ stakes, and suppose agent i has utility u_i satisfying Assumption 1, Assumption 2, and Assumption 5 with*

$$d \leq (M-1)(K-1). \tag{10}$$

Suppose the design matrix (Theorem 10) has maximal rank (defined in Step 3 below; this holds generically by Theorem 10). Then:

(i) The direction of the sieve coefficient vector θ^i is identified: the system of FOC equations determines θ^i up to a positive scalar factor. Equivalently, ψ_i is identified up to $\psi_i \to \alpha\psi_i$ for $\alpha > 0$, and beliefs transform as $p_k^i \to (p_k^i)^\alpha / \sum_j (p_j^i)^\alpha$.

(ii) One auxiliary risk-preference measurement pins α. For instance, an observed value of $\mathrm{RRA}_i(x_0) = -x_0\,\psi_i'(x_0)$ *determines α since $\alpha\psi_i$ implies* $\mathrm{RRA} = \alpha\,\mathrm{RRA}_i(x_0)$. *Given* $\alpha = 1$, *beliefs are uniquely recovered from any stake via* (5)*, and cross-stake consistency provides* $(M-1)(K-1)-d$ *testable overidentifying restrictions.*

No cross-agent common utility restriction is required.

Proof. **Step 1: Additive structure.** Define $w_k^{i,s} := \psi_i(c_s g_k^{i,s})$. The within-stake FOC gives $w_k^{i,s} - w_j^{i,s} = \ln(p_j^i/p_k^i) =: D_{kj}^i$ for all s. Cross-stake consistency gives $w_k^{i,s} - w_k^{i,s'} = \delta_{s,s'}^i$ for all k. Normalizing $w_1^{i,1} = 0$:

$$\psi_i(x_k^{i,s}) = \delta_s^i + D_k^i, \qquad \text{where } x_k^{i,s} := c_s g_k^{i,s}, \tag{11}$$

with $K+M-2$ free parameters $(\delta_2^i, \ldots, \delta_M^i, D_2^i, \ldots, D_K^i)$ and $\delta_1^i = D_1^i = 0$.

Step 2: Sieve substitution. Under Assumption 5: $\sum_{l=1}^d \theta_l^i \phi_l(x_k^{i,s}) = \delta_s^i + D_k^i$—a homogeneous linear system of MK equations in $d+(M-1)+(K-1)$ unknowns $(\theta^i, \delta^i, D^i)$. (Homogeneous because the true solution $(\theta_*^i, \delta_*^i, D_*^i)$ satisfies $\mathbf{A}\xi = 0$, and any scalar multiple $\alpha\xi$ does as well.) Assume the basis functions $\phi_1, \ldots, \phi_d$ are linearly independent of the constant function, so the sieve captures the *shape* of ψ_i while the overall level is distributed between δ^i and D^i.

Step 3: Maximal rank and the scaling ambiguity. The design matrix

$$\mathbf{A}_{(k,s),\cdot} = \big(\phi_1(x_k^{i,s}), \ldots, \phi_d(x_k^{i,s}),\ -\mathbf{e}_s,\ -\mathbf{e}_k\big)$$

has $d+(M-1)+(K-1)$ columns and MK rows. Because the true solution $\xi_* \neq 0$ lies in the null space, $\mathrm{rank}(\mathbf{A}) \leq d+(M-1)+(K-1)-1$. *Maximal rank* means this upper bound is attained, so the null space is exactly one-dimensional: $\{\alpha\,\xi_* : \alpha \in \mathbb{R}\}$. Restricting to $\alpha > 0$ (so ψ_i is strictly decreasing), the sieve coefficients are $\alpha\theta_*^i$, and the implied beliefs are $p_k^i \propto \exp(-\alpha D_k^{i,*}) \propto (p_k^{i,*})^\alpha$. This is the power-scale ambiguity identified in part (i).

Step 4: Generic maximal rank. By Theorem 10 below, maximal rank holds for all but a measure-zero set of evaluation point configurations. Admissible primitives generically produce such configurations because the map from $(\boldsymbol{p}^i, \theta^i) \in \mathrm{int}(\Delta^{K-1}) \times \Theta$ to the evaluation points $\{x_k^{i,s}\}$ is analytic (the optimal report $g_k^{i,s}$ is an analytic function of $(\boldsymbol{p}^i, \theta^i, c_s)$ by the implicit function theorem applied to the FOC). The preimage of the analytic degeneracy set under an analytic map is either everything or measure-zero; since one can exhibit admissible primitives with maximal rank (e.g., $K = 3$, $M = 3$, $d = 2$, $\phi_1(x) = x$, $\phi_2(x) = x^2$, $\boldsymbol{p} = (0.5, 0.3, 0.2)$, $c \in \{1, 3, 8\}$, $u'(x) = e^{-x}/x$), the preimage is measure-zero.

Part (ii). The auxiliary measurement determines α (see theorem statement). With $\alpha = 1$, beliefs are recovered from any stake via (5). The remaining $(M-1)(K-1) - d$ degrees of freedom provide testable overidentifying restrictions. □

Lemma 10 (Generic maximal rank of the design matrix)**.** *Suppose* $\{\phi_l\}_{l=1}^d$ *are analytic functions on* $(0, \infty)$*, linearly independent and linearly independent of the constant function, and* $d \leq (M-1)(K-1)$*. Then* $\mathbf{A}$ *has maximal rank* $d + (M-1) + (K-1) - 1$ *for all but a measure-zero set of evaluation point configurations* $(x_k^s)_{k=1,\dots,K;\, s=1,\dots,M} \in (0, \infty)^{MK}$.

The proof appears in Section A.

4.4 Practical Identification Strategy

The identification results suggest a natural workflow. First, administer the mechanism under $M \geq 2$ stakes. Second, test stake invariance: by Theorem 7, invariance holds iff u_i is CRRA, in which case beliefs follow from (7) given an auxiliary risk measurement. If stake invariance is rejected, the researcher takes the nonparametric path: with $M = 2$ stakes and common utility, Theorem 8 characterizes the power-scale observational equivalence and the role of the anchor; with $M \geq 3$ stakes, Theorem 9 identifies each agent's ψ_i individually up to the same power-scale ambiguity. Every path requires one auxiliary risk-preference measurement per agent (e.g., Holt–Laury) to resolve the power-scale ambiguity (Remark 9).[1]

Remark 8 (Near-CRRA weak identification and routing uncertainty)**.** *Theorem 7 is a sharp population characterization, but finite-sample routing based on stake invariance can be weakly identified near the CRRA boundary. If utility is close to CRRA, cross-stake report differences are small, so rounding and optimization noise can make the invariance statistic difficult to distinguish from zero. In practice, the stake-invariance branch should therefore be treated as a screening heuristic rather than a noiseless classifier: researchers should report the invariance statistic, assess robustness to the routing threshold, and treat near-threshold cases as inconclusive (e.g., report both CRRA and nonparametric recoveries, or default to the individual PMD path when* $M \geq 3$*). The threshold sensitivity results in Section F are informative*

[1] The additive structure of the sieve system (11), where the same latent signal (D_k^i, determined by beliefs) is observed through multiple distortion channels (stakes), parallels Kotlarski-type identification in measurement error models (Kotlarski, 1967); the sieve restriction plays the role of bandwidth.

about aggregate estimator robustness to the routing cutoff, but they are not a formal power analysis of the invariance test.

4.5 Estimation: Penalized Minimum Distance

A natural estimator follows from the constructive proof of Theorem 9. For the true sieve coefficients θ^i, beliefs recovered from each stake via (5) must agree. The *penalized minimum-distance* (PMD) estimator chooses sieve parameters to minimize cross-stake belief disagreement:

$$\hat{\theta}^i = \arg\min_{\theta} \sum_{s<s'} \|\hat{\boldsymbol{p}}^{i,s}(\theta) - \hat{\boldsymbol{p}}^{i,s'}(\theta)\|^2 + \lambda_a \big(R(\theta) - \hat{\rho}_i\big)^2, \tag{12}$$

where $\hat{\boldsymbol{p}}^{i,s}(\theta)$ denotes the beliefs implied by θ at stake s, $R(\theta)$ is the implied relative risk aversion, and $\hat{\rho}_i$ is the observed auxiliary risk measurement. The first term exploits overidentifying restrictions from multiple stakes to identify the shape of ψ_i; the second resolves the power-scale ambiguity (part (i) of Theorem 9) by pinning $\alpha = 1$. Together, the two terms uniquely identify θ^i: as report noise $\sigma \to 0$, the PMD objective is uniquely minimized at the true θ^i and recovered beliefs converge to true beliefs. A full treatment of rates and inference is beyond the scope of this paper; Section 5 provides finite-sample guidance.

Remark 9 (Role of the auxiliary risk measurement)**.** *The power-scale ambiguity in Theorems 8 and 9—the family $(u', \boldsymbol{p}) \to (\beta\,(u')^{\alpha}, \boldsymbol{p}^{\alpha})$—is a fundamental feature of EU-based probability elicitation, not an artifact of the estimation procedure. The multiplicative constant β cancels in the inversion formula, but the exponent α does not: it simultaneously rescales the curvature of utility and the concentration of beliefs, in a way that preserves the observed reports at every stake.*

One auxiliary risk-preference measurement per agent resolves α completely, just as knowing ρ is needed to invert CRRA reports via $p_k \propto g_k^{\rho}$. The nonparametric case generalizes the parametric one: cross-stake variation identifies the shape *of $\psi_i = \ln \circ u_i'$ (i.e., the direction of the sieve coefficients), while the auxiliary measurement pins the* scale.

The PMD estimator (12) *implements this via the RRA anchor term $\lambda_a(R(\theta) - \hat{\rho}_i)^2$. The cross-stake disagreement term in the PMD criterion is uninformative about α and admits a degenerate global minimum at $\theta = 0$ (corresponding to $\alpha \to 0$, which maps all beliefs to uniform). The anchor resolves this degeneracy by pinning $\alpha \approx 1$.*

Simulation evidence shows that the anchor need not be precise. Conditional on a correctly centered anchor, belief recovery is largely insensitive to the anchor weight: $\lambda_a = 0.1$ performs nearly as well as $\lambda_a = 5$, indicating that cross-stake variation provides the primary identifying signal once the degeneracy is broken. Even a rough population-level prior (e.g., RRA =

1.5 *for all agents) prevents collapse, though it does not replicate the precision gains from a correctly centered individual measurement. This robustness is relevant given that risk elicitation instruments are known to be noisy and sometimes context-dependent. Section D provides the full simulation analysis.*

4.6 Point Forecast Overidentification

When agents also report point forecasts under a separate scoring rule (e.g., $S_P(g, \pi) = a \cdot 2^{-|g-\pi|}$), the identified utility provides a testable overidentifying restriction. Under risk aversion, the point forecast targets a "u-weighted quantile" $q_u(F)$—generally not the median. Comparing the observed point forecast with $q_u(\hat{F})$ tests the joint model. Rejection may indicate EU misspecification (e.g., probability weighting), violation of common utility across tasks, or framing effects.

5 Simulation Evidence

This section evaluates the finite-sample performance of the PMD estimator through a Monte Carlo study. The central question is whether structural recovery from multi-stake reports improves on the naive benchmark of using raw reports as beliefs. The appendix provides additional exercises including ablation (Section E), penalty and parameter sensitivity (Section F), point forecast analysis (Section G), and convergence diagnostics (Section H).

5.1 Design

I simulate agents with heterogeneous beliefs and risk preferences. Each agent receives a true belief vector drawn from a Dirichlet distribution over $K = 10$ states and a utility function drawn from one of three DGP classes of increasing difficulty:

- `sieve_exact`: the log marginal utility $\psi_i = \ln u_i'$ lies exactly in the log-polynomial sieve space (80% of agents per replication), with the remaining 20% drawn as pure CRRA.
- `mild_misspec`: utilities come from mixed-CRRA (40%), exponential-power (40%), and pure CRRA (20%) families—none lying exactly in the sieve space, but all reasonably well approximated by a low-dimensional sieve.
- `strong_misspec`: a deliberate stress test with wide-parameter mixed-CRRA utilities (80%) that lie genuinely far from the sieve class, plus 20% pure CRRA.

Table 1: Belief recovery: 500 agents $\times$ 200 replications. L_1 error for stated (raw highest-stake report) and PMD-recovered beliefs, percent reduction, and share of agents individually improving. Standard errors across replications in parentheses.

DGP	Regime	Stated L_1		Recovered L_1		% Red.	Improved
		Mean	(SE)	Mean	(SE)		Share (SE)
Sieve Exact	Common Correct	0.274	(0.005)	0.137	(0.003)	50.2%	0.96 (0.00)
Sieve Exact	Common Wrong	0.264	(0.000)	0.160	(0.001)	39.2%	0.88 (0.00)
Sieve Exact	Individual	0.264	(0.000)	0.128	(0.000)	51.6%	0.97 (0.00)
Mild Misspec.	Common Correct	0.250	(0.010)	0.093	(0.004)	62.8%	0.89 (0.01)
Mild Misspec.	Common Wrong	0.240	(0.000)	0.209	(0.002)	13.0%	0.63 (0.01)
Mild Misspec.	Individual	0.240	(0.000)	0.073	(0.000)	69.6%	0.94 (0.00)
Strong Misspec.	Common Correct	0.159	(0.004)	0.133	(0.005)	16.6%	0.68 (0.02)
Strong Misspec.	Common Wrong	0.164	(0.000)	0.167	(0.004)	-1.4%	0.56 (0.01)
Strong Misspec.	Individual	0.164	(0.000)	0.092	(0.000)	44.1%	0.85 (0.00)

Each DGP is crossed with three estimation regimes: (i) *common utility correctly imposed*, where all agents share one utility and the estimator pools; (ii) *common utility misspecified*, where agents have heterogeneous utilities but the estimator incorrectly pools; and (iii) *individual recovery*, where each agent's utility is estimated from their own reports alone.

Each agent reports at $M = 3$ stakes $c \in \{1, 3, 8\}$. Reports are contaminated by logistic-normal noise (s.d. 0.12) and rounded to a 0.01 grid. Each agent also provides one noisy auxiliary risk-preference measurement (s.d. 0.08 in log-RRA). Recovery uses the PMD estimator (12) with a log-polynomial sieve of dimension $d = 2$ and penalty weight $\lambda_a = 5$. The stated-beliefs baseline is the highest-stake raw report. The core results use $N = 500$ agents and 200 replications; all randomness flows from a single master seed.

5.2 Core Results and Individual Dominance

Table 1 presents the main results across all nine regime–DGP combinations.

The PMD estimator improves belief accuracy in almost every cell. Under `sieve_exact`, recovery reduces L_1 error by 39–52% across regimes, with 88–97% of agents benefiting. Under `mild_misspec`, the individual regime achieves 70% error reduction with 94% of agents improving. Even under `strong_misspec`, the individual regime delivers 44% reduction with 85% of agents improving.

A clear pattern emerges: pooling under the wrong homogeneity assumption is costly (-1% to $+39$%), and even correctly pooled estimation underperforms the individual estimator under strong misspecification (17% vs 44%). The individual regime consistently achieves

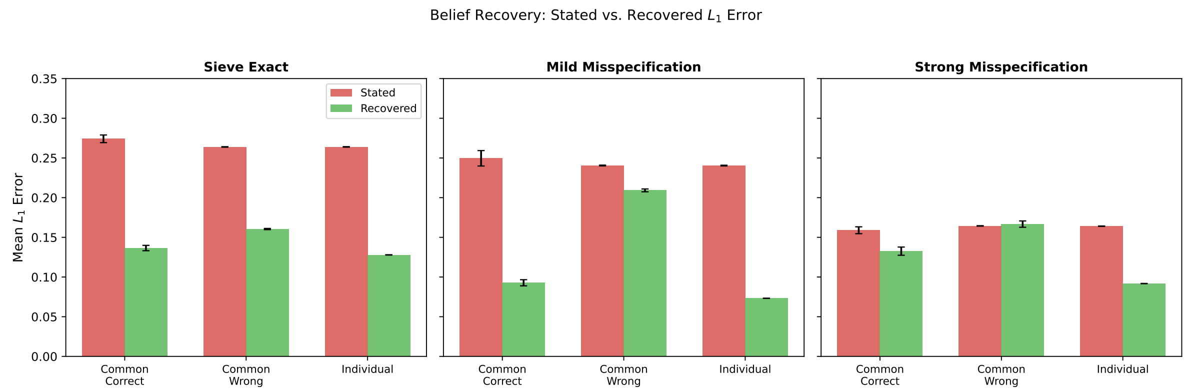

Figure 1: Mean belief L_1 error: stated versus PMD-recovered, by regime and DGP. The PMD estimator reduces error in every cell.

Table 2: Individual estimator robustness: L_1 reduction when preferences are common but estimation is individual, versus correctly pooled and heterogeneous-individual benchmarks. 500 agents $\times$ 200 replications.

DGP	Common Correct	Individual on Common	Individual
Sieve Exact	+50.2%	+50.7%	+51.6%
Mild Misspec.	+62.8%	+72.3%	+69.6%
Strong Misspec.	+16.6%	+42.2%	+44.1%

the best or near-best performance across every DGP. Figure 1 and Figure 2 visualize these results.

An important question is whether the individual estimator sacrifices performance when pooling would be justified. Table 2 addresses this by introducing a fourth regime: *individual-on-common*, where all agents share a common utility but the estimator treats each independently.

The results are decisive. Under `sieve_exact`, individual-on-common (51%) matches both alternatives. Under `mild_misspec`, individual-on-common actually *outperforms* common correct (72% vs. 63%). This reflects misspecified pooling with a finite sieve, not averaging per se: the pooled estimator converges to a single pseudo-true utility that fits the population globally, while belief recovery is evaluated at agent-specific report locations, so separate estimation can fit those local evaluation domains better in this design. Under `strong_misspec`, the advantage is dramatic: 42% vs. 17%. The individual estimator is a dominant strategy: it sacrifices nothing when utility is common and avoids potentially catastrophic pooling bias when it is not.[2]

[2]This dominance is conditional on $M \geq 3$ stakes. With only $M = 2$, the individual estimator is unavailable and pooling under common utility is the only nonparametric option.

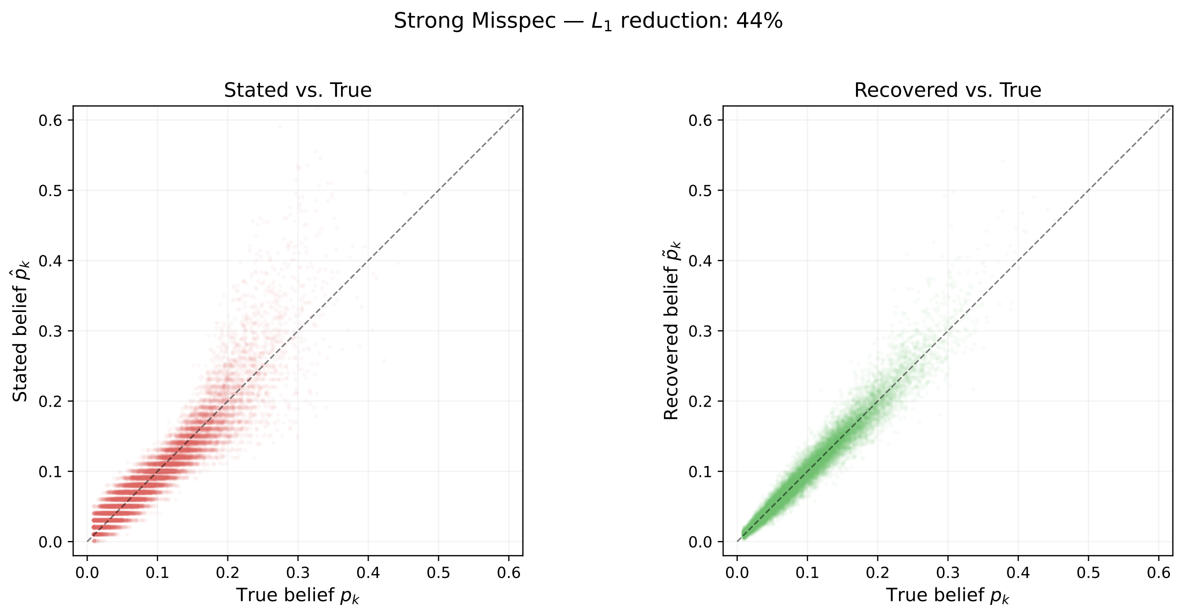

Figure 2: Bin-level scatter: true versus stated (left) and true versus recovered (right) bin probabilities, individual regime, `strong_misspec`. Recovered beliefs cluster tightly around the 45-degree line.

Ablation analysis (Section E) confirms that both components of the PMD objective contribute: the risk anchor does most of the work for easy DGPs, while the cross-stake consistency term is essential under strong misspecification. Penalty weight sensitivity analysis (Section F) shows performance is flat across a 500× range of λ_a. Additional sensitivity analyses (Section F) confirm robustness to sieve dimension ($d \in \{2, 3, 4\}$), number of bins ($K \in \{5, 10, 20\}$), anchor noise (s.d. up to 0.25), heaping noise (focal-point rounding), and a targeted log-spline basis substitution in flagship PMD designs. The CRRA routing-threshold sensitivity reported in Section F should be read as a tuning-robustness check for aggregate recovery, not as a formal power analysis of stake-invariance routing near the CRRA boundary (see Remark 8); Section F also reports explicit feasible-versus-oracle routing comparisons and a dedicated near-CRRA routing exercise.

5.3 Out-of-Sample Validation

The most demanding test of whether the estimator learns genuine utility structure is out-of-sample prediction. I use four stakes $c \in \{1, 3, 5, 8\}$, hold one out, and estimate utility from the remaining three. I then predict beliefs at the held-out stake. Since the held-out data played no role in estimation, positive performance constitutes evidence of genuine learning.

Under interpolation (held-out stake within the training range), out-of-sample reductions

Table 3: Out-of-sample validation: fit with 3 of 4 stakes, predict at the held-out stake. "IS" is in-sample L_1 reduction; "OOS" is out-of-sample. 500 agents $\times$ 50 replications.

DGP	Held Out	Type	IS % Red.	OOS % Red.	OOS Impr.
Sieve Exact	$c = 1$	Extrap.	51.2%	8.3%	0.64
Sieve Exact	$c = 3$	**Interp.**	51.5%	21.2%	0.74
Sieve Exact	$c = 5$	**Interp.**	51.5%	24.9%	0.79
Sieve Exact	$c = 8$	Extrap.	51.3%	4.8%	0.65
Mild Misspec.	$c = 1$	Extrap.	69.1%	61.8%	0.87
Mild Misspec.	$c = 3$	**Interp.**	69.8%	62.8%	0.88
Mild Misspec.	$c = 5$	**Interp.**	69.8%	61.7%	0.88
Mild Misspec.	$c = 8$	Extrap.	70.9%	46.0%	0.73
Strong Misspec.	$c = 1$	Extrap.	47.0%	26.7%	0.78
Strong Misspec.	$c = 3$	**Interp.**	45.6%	29.8%	0.79
Strong Misspec.	$c = 5$	**Interp.**	44.1%	25.3%	0.73
Strong Misspec.	$c = 8$	Extrap.	47.0%	-13.7%	0.45

are positive across all DGPs, including strong misspecification (+25–30%). Under mild misspecification, OOS performance nearly matches in-sample (+62% vs. +70%). Extrapolation to stakes beyond the training range is less reliable, as expected. The practical implication: training stakes should span the range at which belief recovery is desired. Table 3 and Figure 3 report the full results. Derived scalar forecasts (probability-weighted means) show analogous improvements (Section G).

Convergence exercises (Section H) reveal an important additional finding: the individual estimator's performance is stable across sample sizes $N \in \{50, \ldots, 2000\}$, while pooled estimators with fixed sieve dimension degrade at large N under misspecification—converging precisely to the wrong answer. This further strengthens the case for the individual approach as the robust default.

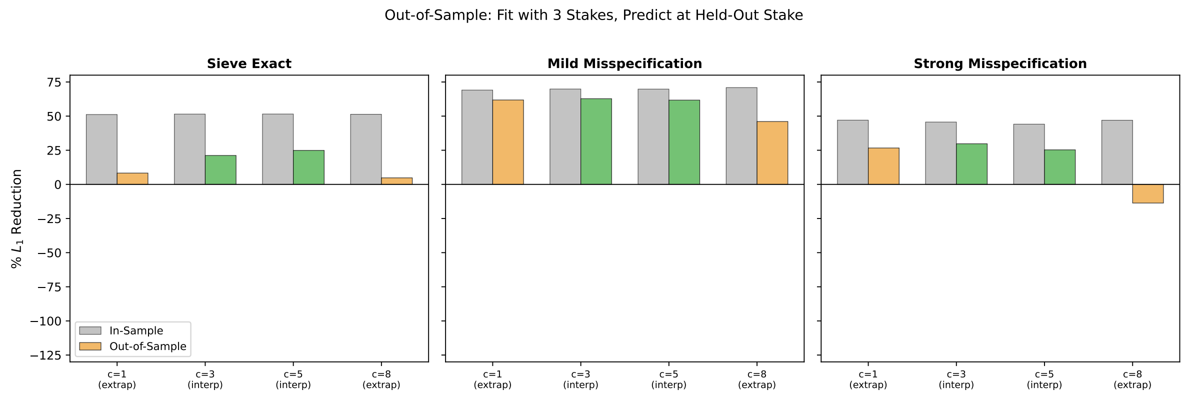

Figure 3: Out-of-sample validation. Gray bars: in-sample L_1 reduction. Colored bars: out-of-sample reduction (green = interpolation, orange = extrapolation). The estimator learns utility structure that transfers to unseen stakes within the training range.

5.4 Recovering Point Forecasts From Bins

A common use of binned probability elicitations in applied work is to construct scalar summary measures of expectations, typically the probability-weighted mean of bin midpoints. This practice is standard in analyses of household surveys such as the SCE, HRS, and SEE, where researchers convert binned inflation, income, or survival expectations into point forecasts for regressions, forecast evaluation, and policy analysis. Because the distortion from risk preferences operates on the entire reported distribution, it propagates directly into any derived scalar statistic. The structural recovery developed here corrects the full distribution before aggregation, so the resulting point forecasts inherit the improvement. Here I show how improving belief recovery from binned forecasts in the simulations translates into better recovered summary statistics.

Table 4: Point forecast exercise: mean absolute error (MAE) of stated and recovered point forecasts.

DGP	MAE Stated	MAE Recovered	% Red.	Improved
Sieve Exact	0.0262	0.0134	48.9%	0.72
Strong Misspec.	0.0175	0.0098	44.0%	0.67

This is a standard nonparametric approach to summarizing binned elicitations; in applied work, researchers typically either compute probability-weighted bin midpoint means directly or fit a parametric distribution (most commonly a generalized beta, following Engelberg et al. (2009)) and extract its mean. Engelberg et al. (2009) show that the two approaches yield nearly identical central tendency measures for the Survey of Professional Forecasters, and the SCE literature has adopted this methodology as standard practice (Armantier et al., 2017).

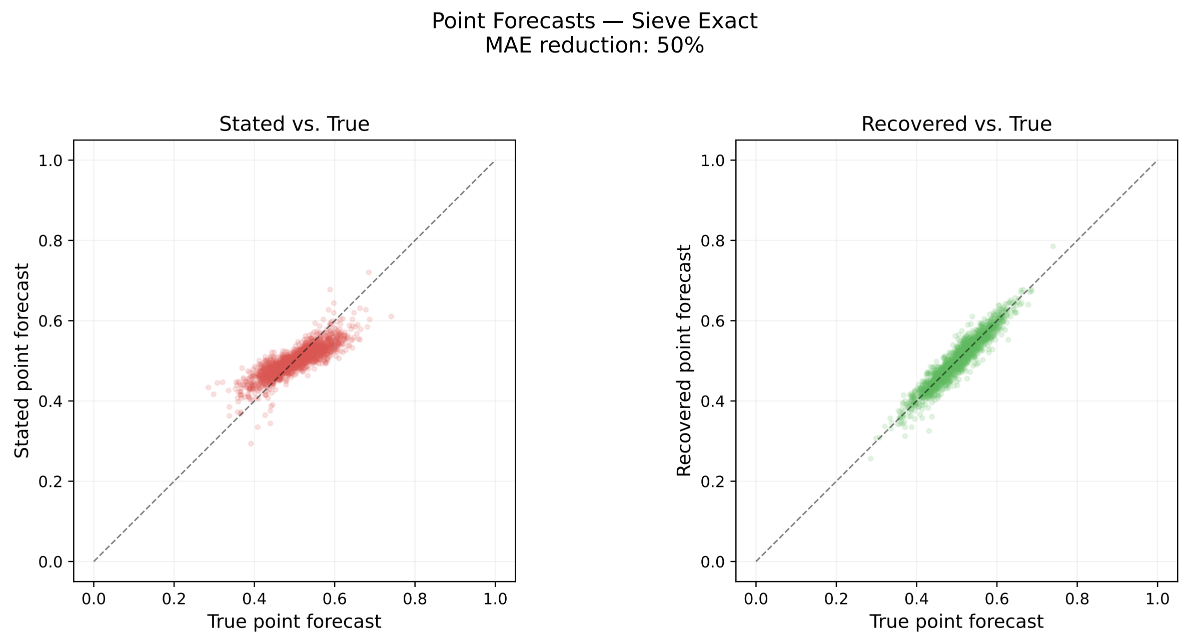

Figure 4: Point forecast scatter: `sieve_exact`. Left: stated vs. true. Right: recovered vs. true.

The midpoint mean used here is the simpler of the two and, because the simulation uses evenly spaced bins with known endpoints, avoids any additional distributional assumption.

Table 4 shows that the distributional gains carry through to the scalar summaries used in applied work. Under `sieve_exact`, PMD reduces point-forecast MAE from 0.0262 to 0.0134 (48.9%), and 72% of agents' point forecasts move closer to the truth. Even under `strong_misspec`, recovered point forecasts remain substantially more accurate than stated forecasts: MAE falls from 0.0175 to 0.0098 (44.0%), with 67% of agents improving. Figure 4 shows the same mechanism graphically in the `sieve_exact` design: correcting the full reported distribution before aggregation moves derived point forecasts closer to the 45-degree line.

5.5 Practical Guidance

The simulation evidence yields four operational recommendations:

1. *Use the individual estimator as default.* It matches or outperforms correctly pooled estimation across every DGP, sacrifices nothing when preferences are common, and is immune to large-N degradation.

2. *Administer at least three stakes spanning the range of interest.* Three stakes are the

theoretical minimum for individual identification; the out-of-sample results show reliable prediction within the training range but unreliable extrapolation.

3. *Include a risk anchor, but it need not be precise.* The RRA anchor resolves the power-scale ambiguity identified in Theorem 9: cross-stake variation identifies the shape of utility but not its scale, so one auxiliary risk measurement per agent is needed (Remark 9). Anchor centering matters more than anchor weight—a correctly centered anchor at $\lambda_a = 0.1$ performs nearly as well as $\lambda_a = 5$, indicating that the cross-stake variation provides the primary identifying signal once the scale ambiguity is broken. Standard incentivized risk tasks (e.g., Holt–Laury (Holt and Laury, 2002), Eckel–Grossman (Eckel and Grossman, 2002, 2008), or bomb risk elicitation (Crosetto and Filippin, 2013)) provide sufficient precision; even a rough population-level prior prevents degeneracy. See Section D for the full sensitivity analysis.

4. *Results are robust to tuning and noise structure.* The penalty weight can vary across a $500\times$ range with negligible effect, and gains persist under heaping noise and across sieve dimensions $d \in \{2, 3, 4\}$.

6 Robustness and Extensions

6.1 Rank-Dependent Utility

Under rank-dependent utility (RDU) (Quiggin, 1982; Tversky and Kahneman, 1992), the agent evaluates a report using decision weights $w_k(\boldsymbol{p})$ generated by a probability weighting function. Conditional on a fixed payoff ranking (which is generic—ties arise on measure-zero sets), the RDU problem is formally identical to EU with p_k replaced by $w_k(\boldsymbol{p})$, yielding the FOC

$$w_k(\boldsymbol{p})\, u'(c\, g_k^*) = \lambda, \qquad k \in S(\boldsymbol{p}). \tag{13}$$

The implications for the main results are as follows. Inversion (5) recovers decision weights $w_k(\boldsymbol{p})$ rather than beliefs p_k; recovering beliefs requires additional restrictions on w. Stake invariance continues to characterize CRRA on any domain where the optimal ranking is constant across stakes, since cross-stake FOC ratios eliminate decision weights:

$$\frac{u'(c_1\, g_j^{i,1})}{u'(c_1\, g_k^{i,1})} = \frac{w_j(\boldsymbol{p}^i)}{w_k(\boldsymbol{p}^i)} = \frac{u'(c_2\, g_j^{i,2})}{u'(c_2\, g_k^{i,2})}.$$

Hence the identification results for u' extend to RDU on regions without rank reversals, which can be tested directly in the data. Under RDU, truthful reporting for all $(\boldsymbol{p}, c)$ requires both

$u(x) = A \ln x + B$ and $w = \mathrm{id}$.

6.2 Ambiguity Aversion

Under maxmin EU (Gilboa and Schmeidler, 1989), the agent hedges against the worst-case prior, compressing reports similarly to risk aversion. The inversion formula recovers the worst-case prior, not the center of the prior set. Systematic patterns—particularly if recovered "beliefs" depend on the stake after accounting for utility—may indicate ambiguity aversion.

6.3 Reported Zeros, Rounding, and Background Wealth

Under the baseline model, $g_k^* = 0$ implies $p_k = 0$. In practice, reported zeros often reflect rounding (Manski and Molinari, 2010). ε-smoothing—replacing $\tilde{g}_k$ with $\max\{\tilde{g}_k, \varepsilon\} / \sum_j \max\{\tilde{g}_j, \varepsilon\}$—handles this; the simulation uses $\varepsilon = 0.01$.

Extending the model to $u(w + c\, g_k)$ with background wealth $w > 0$ permits corner solutions at bins with small but positive beliefs, since $u'(w) < \infty$. If w is known, the identification results apply directly with shifted evaluation points. If w is unknown but common, it adds one nuisance parameter to the sieve ($d + 1 \leq (M - 1)(K - 1)$, easily satisfied for $K \geq 10$). If w varies across agents, an auxiliary wealth measurement or a parametric restriction is needed. In experimental settings, "narrow bracketing" instructions mitigate background wealth effects (Harrison et al., 2017).

Note that the inversion procedure does not alter agents' incentives: under PtP, the agent's payoff depends only on the realized bin and reported probability, not the researcher's subsequent use of the data. The overidentifying restrictions from multiple stakes provide a diagnostic for noisy optimization and rounding severity.

They also provide a falsification check for the stake-exogeneity assumption (beliefs fixed across stakes). With $M \geq 3$ stakes, the model implies that a single latent belief vector and utility index must jointly rationalize all stake-specific reports; equivalently, after anchoring the power scale, recovered beliefs from different stakes should agree up to noise. In practice, one can report the PMD overidentification residuals (or leave-one-stake-out prediction errors) and examine whether they load systematically on stake level or stake order. Persistent, stake-directional residual patterns after accounting for rounding/noise are evidence against the exclusion restriction, suggesting stake-induced learning, attention, or effort responses rather than pure preference-driven distortion.

Administering multiple stakes raises standard experimental concerns (order effects, learning), which are addressed by randomized stake ordering and temporal separation; the residual diagnostics above provide an ex post complement to these design safeguards.

7 Conclusion

The pay-the-probability rule is not incentive compatible under standard risk-neutrality assumptions, but under risk aversion its distortions are structured and invertible. This paper exploits this structure: stake variation serves as an exclusion restriction that identifies utility structure up to a power-scale ambiguity, and inversion plus one auxiliary risk-preference measurement recovers true beliefs. The practical recommendation is clear: administer the PtP mechanism at three stake sizes, collect a standard risk-preference measurement, and apply the individual PMD estimator. This reduces belief error by 44–70% in Monte Carlo exercises, requires no assumptions about utility homogeneity, and learns genuine utility structure as confirmed by out-of-sample validation. The most immediate application is to the SCE, HRS, and similar household expectation surveys that already use binned probability elicitation and currently treat raw reports as beliefs.

Three directions for future work are immediate. First, developing the formal asymptotic theory for the PMD estimator—rates of convergence, optimal penalty selection, and inference. Second, developing joint identification of utility u and probability weighting w from richer experimental designs (Section 6.1). Third, experimental validation: implementing the three-stake design and testing whether recovered beliefs and preferences are consistent with auxiliary measures.

A Deferred Proofs

A.1 Proof of Theorem 1, Part (i): Full Details

Support recovery: $p_k = 0 \Rightarrow g_k^* = 0$. If $p_k = 0$, the term $p_k u(cg_k)$ vanishes for all g_k. Any mass allocated to bins with $p_k = 0$ reduces mass available to bins in $S(\boldsymbol{p})$ without increasing V. Hence $g_k^* = 0$.

Interiority: $p_k > 0 \Rightarrow g_k^* > 0$. Consider two cases. If $u(0^+) = -\infty$, then $V(\boldsymbol{g}) \to -\infty$ whenever $g_k \to 0$ for any $k \in S(\boldsymbol{p})$, so the optimum must have $g_k^* > 0$. If $u(0^+) > -\infty$ but $u'(0^+) = +\infty$ (Inada), then for $j \in S(\boldsymbol{p})$ with $g_j = 0$, shifting ε mass from any k with $g_k > 0$ yields directional derivative $c[p_j u'(0^+) - p_k u'(cg_k)] = +\infty$, so $g_j = 0$ cannot be optimal.

Existence and uniqueness. The support face is compact; V is continuous, so a maximizer exists (Weierstrass). On the face, V is strictly concave ($u'' < 0$), yielding uniqueness.

A.2 Proof of Lemma 4: Local Invertibility

Fix $(\boldsymbol{p}, c) \in \operatorname{int}(\Delta^{K-1}) \times (0, \infty)$ and let $(\boldsymbol{g}^*, \lambda)$ solve (4). Define $F_k(\boldsymbol{g}, \lambda, \boldsymbol{p}) := p_k\, u'(c\, g_k) - \lambda$ for $k = 1, \ldots, K$ and $F_{K+1}(\boldsymbol{g}) := \sum_k g_k - 1$. The Jacobian $J_{(\boldsymbol{g},\lambda)}$ is a bordered diagonal matrix with diagonal entries $p_k c\, u''(cg_k^*)$ and border $(-1, \ldots, -1; 1, \ldots, 1)$. Setting (h, η) in the null space: $h_k = \eta/[p_k c\, u''(cg_k^*)]$ and $\sum_k h_k = 0$. Since all denominators are negative, $\eta = 0$ and $h = 0$. Nonsingularity of J and the implicit function theorem give $(\boldsymbol{g}^*, \lambda)$ locally C^1 in $\boldsymbol{p}$.

Restricting to the simplex tangent space T: if $d\boldsymbol{g} = 0$ then the FOC yields $dp_k = d\lambda/u'(cg_k^*)$, so $\sum_k dp_k = d\lambda \sum_k 1/u'(cg_k^*) = 0$, giving $d\lambda = 0$ and $d\boldsymbol{p} = 0$. The map has rank $K - 1$.

A.3 Full Converse Characterization for Theorem 8

The main text proves the constructive direction:

$$(u', p) \mapsto (\widetilde{u}', \widetilde{p}) = \big(\beta(u')^\alpha,\ p^\alpha/\|p^\alpha\|_1\big)$$

preserves the two-stake FOCs. This subsection proves the converse under an explicit overlap-normalization condition.

Assumption 6 (Overlap-normalization (ON))**.** *Let*

$$J := \psi([a, b]), \qquad \psi := \ln \circ u', \qquad \widetilde{\psi} := \ln \circ \widetilde{u}'.$$

For each agent i, define

$$\Delta_i := \delta_{i,2} - \delta_{i,1}, \qquad \widetilde{\Delta}_i := \widetilde{\delta}_{i,2} - \widetilde{\delta}_{i,1}, \qquad T_i := \{\delta_{i,1} + D_{i,k} : k \in S(p_i)\} \subset J \cap (J - \Delta_i),$$

where $\delta_{i,s} := \ln \lambda_{i,s}$, $D_{i,k} := -\ln p_{i,k}$, and similarly $\widetilde{\delta}_{i,s} := \ln \widetilde{\lambda}_{i,s}$, $\widetilde{D}_{i,k} := -\ln \widetilde{p}_{i,k}$.

Assume:

1. *$\mathcal{D} := \{\Delta_i : i \in \mathcal{I}\}$ is dense in a non-degenerate interval containing 0.*
2. *If $\Delta_i = \Delta_j$, then $\widetilde{\Delta}_i = \widetilde{\Delta}_j$.*
3. *For each $\Delta \in \mathcal{D}$, the set*
$$T(\Delta) := \bigcup_{i : \Delta_i = \Delta} T_i$$
is dense in $J \cap (J - \Delta)$.

Proposition 11 (Converse power-scale characterization)**.** *Suppose Assumptions 1, 3, 4, and ON hold. Let* $(u, \{p_i\}_{i\in\mathcal{I}})$ *and* $(\widetilde{u}, \{\widetilde{p}_i\}_{i\in\mathcal{I}})$ *be two rationalizations of the same two-stake reports* $\{g_{i,s}\}$*. Then there exist constants* $\alpha > 0$*,* $\beta > 0$ *such that on* $[a, b]$*,*

$$\widetilde{u}'(x) = \beta\,[u'(x)]^{\alpha},$$

and for each i*,*

$$\widetilde{p}_{i,k} = \frac{(p_{i,k})^{\alpha}}{\sum_j (p_{i,j})^{\alpha}}.$$

Hence the observational equivalence class is exactly the power-scale class.

Proof. Fix any rationalization. For each i, s, k, with $x_{i,s,k} := c_s g_{i,s,k}$, the FOCs are

$$p_{i,k}u'(x_{i,s,k}) = \lambda_{i,s}, \qquad \widetilde{p}_{i,k}\widetilde{u}'(x_{i,s,k}) = \widetilde{\lambda}_{i,s}.$$

Taking logs:

$$\psi(x_{i,s,k}) = \delta_{i,s} + D_{i,k}, \qquad \widetilde{\psi}(x_{i,s,k}) = \widetilde{\delta}_{i,s} + \widetilde{D}_{i,k}.$$

Define $h := \widetilde{\psi} \circ \psi^{-1}$ on J. Then

$$h(\delta_{i,s} + D_{i,k}) = \widetilde{\delta}_{i,s} + \widetilde{D}_{i,k}.$$

Subtract $s = 1$ from $s = 2$:

$$h(t + \Delta_i) - h(t) = \widetilde{\Delta}_i, \qquad t \in T_i. \tag{A.3.1}$$

Now fix $\Delta \in \mathcal{D}$. By ON(2), $\widetilde{\Delta}_i$ is common across all i with $\Delta_i = \Delta$; call this common value $\kappa(\Delta)$. Then (A.3.1) implies

$$h(t + \Delta) - h(t) = \kappa(\Delta), \qquad t \in T(\Delta).$$

By ON(3), $T(\Delta)$ is dense in $J \cap (J - \Delta)$, and h is continuous, so

$$h(t + \Delta) - h(t) = \kappa(\Delta), \qquad \forall t \in J \cap (J - \Delta). \tag{A.3.2}$$

Take $\Delta_1, \Delta_2 \in \mathcal{D}$ and t such that all terms are defined. Using (A.3.2):

$$\begin{aligned} h(t + \Delta_1 + \Delta_2) - h(t) &= [h(t + \Delta_1 + \Delta_2) - h(t + \Delta_1)] + [h(t + \Delta_1) - h(t)] \\ &= \kappa(\Delta_2) + \kappa(\Delta_1). \end{aligned}$$

Hence $\kappa(\Delta_1+\Delta_2)=\kappa(\Delta_1)+\kappa(\Delta_2)$ whenever $\Delta_1,\Delta_2,\Delta_1+\Delta_2\in\mathcal{D}$. Also κ is continuous (on $\mathcal{D}$, via $\kappa(\Delta)=h(t+\Delta)-h(t)$ for fixed interior t). Therefore κ is linear:

$$\kappa(\Delta)=\alpha\Delta \quad \text{on } \mathcal{D}$$

for some $\alpha\in\mathbb{R}$.

Substitute into (A.3.2):

$$h(t+\Delta)-h(t)=\alpha\Delta, \qquad \Delta\in\mathcal{D},\ t\in J\cap(J-\Delta). \tag{A.3.3}$$

For fixed t, the map $\Delta\mapsto h(t+\Delta)-h(t)-\alpha\Delta$ is continuous and vanishes on dense $\mathcal{D}$ (ON(1)); thus it vanishes for all admissible Δ. So for all $t,t'\in J$:

$$h(t')-h(t)=\alpha(t'-t).$$

Hence h is affine on J:

$$h(t)=\alpha t+\beta_0.$$

Because $u',\widetilde{u}'>0$ and both are strictly decreasing, $\psi,\widetilde{\psi}$ are strictly decreasing; therefore $h=\widetilde{\psi}\circ\psi^{-1}$ is strictly increasing, so $\alpha>0$.

Thus on $[a,b]$:

$$\widetilde{\psi}(x)=\alpha\psi(x)+\beta_0 \quad\Longleftrightarrow\quad \widetilde{u}'(x)=e^{\beta_0}[u'(x)]^{\alpha}.$$

Set $\beta:=e^{\beta_0}>0$. This proves the utility part.

For beliefs, from the log-FOCs:

$$p_{i,k}=e^{\delta_{i,s}}e^{-\psi(x_{i,s,k})}, \qquad \widetilde{p}_{i,k}=e^{\widetilde{\delta}_{i,s}}e^{-\widetilde{\psi}(x_{i,s,k})}.$$

Using $\widetilde{\psi}=\alpha\psi+\beta_0$:

$$\widetilde{p}_{i,k}=e^{\widetilde{\delta}_{i,s}-\beta_0-\alpha\delta_{i,s}}(p_{i,k})^{\alpha}=:C_{i,s}(p_{i,k})^{\alpha}.$$

Normalizing over k (since $\sum_k\widetilde{p}_{i,k}=1$):

$$\widetilde{p}_{i,k}=\frac{(p_{i,k})^{\alpha}}{\sum_j(p_{i,j})^{\alpha}}.$$

So the converse is exactly the power-scale class. $\square$

Anchor implication. If one observes $RRA(x_0) = -x_0 u''(x_0)/u'(x_0)$ at any $x_0 \in [a, b]$, then

$$\widetilde{RRA}(x_0) = \alpha\, RRA(x_0),$$

so α is pinned; β cancels in inversion formula (5).

A.3 Proof of Lemma 10: Generic Maximal Rank

Write $\mathbf{A} = [\Phi \mid -S \mid -B]$ where Φ is the $MK \times d$ matrix with $\Phi_{(k,s),l} = \phi_l(x_k^s)$, S is the $MK \times (M-1)$ stake-indicator matrix with $S_{(k,s),s'} = \mathbf{1}\{s = s'+1\}$, and B is the $MK \times (K-1)$ bin-indicator matrix with $B_{(k,s),k'} = \mathbf{1}\{k = k'+1\}$.

Step 1: Upper bound on rank. The vector $\xi_0 := (0, \ldots, 0,\ 1, \ldots, 1,\ -1, \ldots, -1)$ with $\theta = 0$, $\delta_s = 1$ for all $s \geq 2$, and $D_k = -1$ for all $k \geq 2$ does *not* lie in the null space in general (the equation $\Phi \cdot 0 = S\mathbf{1} + B\mathbf{1}$ would require the indicators to sum to zero at each row, which they do not). Instead, the rank bound follows because any nonzero solution ξ_* to the sieve system lies in the null space. Hence $\text{rank}(\mathbf{A}) \leq d + (M-1) + (K-1) - 1$.

Step 2: Existence of a maximal-rank configuration. Choose MK distinct points $x_1 < x_2 < \cdots < x_{MK}$ in $(0, \infty)$ and a polynomial basis $\phi_l(x) = x^l$ for $l = 1, \ldots, d$ (excluding the constant, consistent with the theorem's convention). The sieve block Φ is then a Vandermonde-type matrix with distinct nodes, which has rank $\min(MK, d) = d$. The indicator columns S and B are $\{0, 1\}$-valued with disjoint support patterns, so $[S \mid B]$ has rank $(M-1)+(K-1)$. Because $\phi_1, \ldots, \phi_d$ are linearly independent of the constant function, the columns of Φ are not in the span of $[S \mid B]$ generically. One verifies directly (e.g., for $K = 3$, $M = 3$, $d = 2$) that $\text{rank}(\mathbf{A}) = d + (M-1) + (K-1) - 1$ at the chosen configuration.

Step 3: Generic extension. Each maximal minor of $\mathbf{A}$ is an analytic function of the MK evaluation points $(x_1, \ldots, x_{MK}) \in (0, \infty)^{MK}$. The rank condition $\text{rank}(\mathbf{A}) < d + (M-1) + (K-1) - 1$ requires *every* maximal minor of size $d + (M-1) + (K-1) - 1$ to vanish. By Step 2, at least one such minor is nonzero at the Vandermonde configuration. An analytic function on $(0, \infty)^{MK}$ that is not identically zero vanishes on a set of Lebesgue measure zero. Therefore $\mathbf{A}$ has maximal rank for all but a measure-zero set of evaluation point configurations.

A.4 Binary Elicitation and Non-Uniqueness

When $K = 2$, the truthful reporting condition becomes $p\, u'(cp) = (1-p)\, u'(c(1-p))$ for all $p \in (0, 1)$. Defining $h(p) := p\, u'(cp)$, this requires $h(p) = h(1-p)$—symmetry about $1/2$. The general solution is $u'(x) = (A/x) \cdot \tilde{h}(x/c)$ where $\tilde{h}$ is any admissible symmetric function, yielding a family parameterized by an arbitrary function rather than a single utility. Logarithmic utility corresponds to $\tilde{h} \equiv 1$. Binary elicitation therefore does not generate sufficient variation to uniquely identify utility curvature.

B Extension to General Separable Payoff Maps

The main text focuses on the pay-the-probability (PtP) rule, where the payoff in bin k is $c \cdot g_k$. All results extend naturally to the class of *separable bin-elicitation mechanisms*

$$S(\boldsymbol{g}, \pi) = c \cdot s(g_k) \quad \text{if } \pi \in b_k, \tag{14}$$

where $s : [0,1] \to [0,\infty)$ is a payoff map satisfying: (i) $s \in C^2(0,1)$ with $s(0) = 0$; (ii) $s'(x) > 0$ on $(0,1)$; and (iii) the composite $x \mapsto u(c\, s(x))$ is strictly concave on $(0,1)$. Condition (iii) holds automatically when s is concave, and more generally when the concavity of u dominates: $u''(c\, s(x))\,(s'(x))^2 + u'(c\, s(x))\, s''(x) < 0$. PtP corresponds to $s = \mathrm{id}$.

B.1 General FOC and Inversion Formula

Under general s, the agent's problem is $\max_{\boldsymbol{g} \in \Delta^{K-1}} \sum_k p_k\, u(c\, s(g_k))$. The FOC on the support becomes

$$p_k\, u'(c\, s(g_k^*))\, s'(g_k^*) = \lambda, \qquad k \in S(\boldsymbol{p}), \tag{15}$$

and the inversion formula generalizes to

$$p_k = \frac{1/[u'(c\, s(g_k^*))\, s'(g_k^*)]}{\sum_{j \in S(\boldsymbol{p})} 1/[u'(c\, s(g_j^*))\, s'(g_j^*)]}, \qquad k \in S(\boldsymbol{p}). \tag{16}$$

Since s is a known design choice of the experimenter, this adds no unknown quantities relative to the PtP case.

B.2 Truthful Reporting under General Payoff Maps

Under general s, the truthful reporting characterization (Theorem 2) extends to a *joint* characterization of u and s. The optimal report satisfies $\boldsymbol{g}^* = \boldsymbol{p}$ for all interior beliefs and all $c > 0$ if and only if: (a) u is CRRA with parameter $\rho > 0$, and (b) s satisfies the ODE

$$x\, s'(x)\, s(x)^{-\rho} = \beta, \qquad x \in (0,1), \tag{17}$$

for some $\beta > 0$. When $\rho = 1$, the unique solution with $s(0) = 0$ is $s(x) = D\,x^{\beta}$ (a power function). Under PtP, the payoff map is fixed at $s(x) = x$ (i.e., $D = 1$, $\beta = 1$), so only $\rho = 1$ (log utility) achieves truthfulness—confirming Theorem 2 as a special case. When $\rho \neq 1$, integrating (17) yields

$$s(x)^{1-\rho} = (1 - \rho)\beta \ln x + C.$$

This implies an important boundary qualification under the maintained design condition $s(0) = 0$: for $\rho > 1$, the ODE admits a one-parameter family of increasing solutions on $(0, 1)$ consistent with the boundary behavior at zero, but for $\rho < 1$ no real-valued, nonnegative, strictly increasing solution exists on the full interval $(0, 1)$, because the right-hand side becomes negative for sufficiently small x. Thus non-log CRRA truthful implementation is feasible on the full domain only for $\rho > 1$ (or $\rho = 1$), unless the relevant reporting domain is bounded away from zero or the boundary condition is relaxed.

B.3 Stake Invariance and Identification

The stake invariance characterization (Theorem 7) is *independent* of the payoff map: under any s satisfying the regularity conditions above, stake invariance holds if and only if u is CRRA. The proof carries over verbatim, since the s' terms cancel in the cross-stake FOC ratios when $\boldsymbol{g}^*$ is the same under both stakes.

The population two-stake power-scale equivalence result and the individual identification result (Theorems 8 and 9) also extend. The evaluation points become $x_k^{i,s} = c_s\, s(g_k^{i,s})$ rather than $c_s\, g_k^{i,s}$. Taking logs of the general FOC (15): $\psi_i(x_k^{i,s}) = \ln \lambda^{i,s} - \ln p_k^i - \ln s'(g_k^{i,s})$. Since s is known, $\ln s'(g_k^{i,s})$ is a known offset, and the sieve system retains the additive structure after absorbing it: $\psi_i(x_k^{i,s}) + \ln s'(g_k^{i,s}) = \delta_s^i + D_k^i$. The left-hand side involves only known quantities (the sieve function at known evaluation points, plus a known correction), so the identification argument proceeds as before. Since s is known and strictly monotone, the evaluation points are distinct whenever the reports are distinct, and the connectivity and generic rank conditions hold under the same assumptions.

C Finite Composite Marginal Utility at Zero

The baseline model (Assumption 1) imposes the Inada condition $u'(0^+) = +\infty$, which ensures the clean biconditional $g_k^* = 0 \Leftrightarrow p_k = 0$. Relaxing this to $u'(0^+) < \infty$ (e.g., $u(x) = \ln(1+x)$ with $u'(0^+) = 1$) permits corner solutions at bins with small but positive beliefs.

Under a general payoff map s, the relevant boundary condition is the *composite Inada condition*: $\lim_{x \to 0^+} u'(c\, s(x))\, s'(x) = +\infty$. When this fails, the KKT condition for an inactive bin becomes $p_k\, u'(c\, s(0^+))\, s'(0^+) \le \lambda$, so bin k is rationally shut down whenever $p_k \le \lambda/[u'(c\, s(0^+))\, s'(0^+)]$—the agent assigns positive belief but reports zero because the marginal value of allocating probability mass to a low-probability bin does not justify the cost. Under PtP, this simplifies to $p_k \le \lambda/u'(0^+)$.

Identification and inversion proceed on the active set $\{k : g_k^* > 0\}$ via the same FOC (4). Inactive bins yield the one-sided bound $p_k \le \lambda/[u'(c\, s(0^+))\, s'(0^+)]$ rather than point identification. The sieve identification result (Theorem 9) applies to the active bins without modification. The threshold for corner solutions depends on c (through λ), so varying the stake shifts the active set: bins that are inactive under a small stake may become active under a larger stake, providing additional identifying information at the boundary.

D Anchor Sensitivity Analysis

This appendix provides the simulation evidence underlying Remark 9. The goal is to understand what the RRA anchor contributes to belief recovery: how much precision does the auxiliary risk measurement need to carry, given that the cross-stake variation identifies the shape of utility?

D.1 Setup

All exercises use the individual PMD estimator (12) under the `strong_misspec` DGP (mixed-CRRA utilities), which represents the most demanding environment. Agents have heterogeneous utility $u_i'(x) = x^{-\rho_i} + a_i x^{-\gamma_i}$ with $\rho_i \sim U[0.8, 2.5]$, $\gamma_i \sim U[0.2, 0.8]$, $a_i \sim U[0.3, 2.0]$. Beliefs are drawn from a Dirichlet distribution over $K = 10$ bins. Reports are observed at stakes $c \in \{1, 3, 8\}$ with logistic-normal noise (s.d. 0.12) and rounded to a 0.01 grid. The anchor observation is the true relative risk aversion evaluated at the median consumption point, observed with log-normal noise (s.d. 0.08). The sieve dimension is $d = 2$ (log-polynomial basis). Performance is measured by L_1 belief error $= \sum_k |p_k - \hat{p}_k|$. This appendix reports absolute L_1 levels from a targeted anchor-comparison Monte Carlo (separate from the production 'core'/'ablation' tables, which report percent reductions under a different simulation pipeline), so the numeric levels here should not be directly compared to Tables 1 and 5.

D.2 The degenerate minimum at $\theta = 0$

The cross-stake disagreement term in (12), $\sum_{s<s'} \|\hat{\boldsymbol{p}}^{i,s}(\theta) - \hat{\boldsymbol{p}}^{i,s'}(\theta)\|^2$, measures how well beliefs recovered at different stakes agree. Ideally, at the true θ, beliefs from all stakes coincide. However, this criterion has a trivial global minimum at $\theta = 0$: if $\psi_i \equiv 0$, then u_i' is

constant (risk neutrality), and the inversion formula yields $\hat{p}_k^{(s)} = 1/K$ (uniform beliefs) at every stake. Uniform beliefs agree perfectly across stakes, so the cross-stake disagreement is exactly zero—but the recovered beliefs are meaningless.

This reflects the power-scale ambiguity identified in Theorem 9(i): the cross-stake disagreement is zero for any scalar multiple $\alpha\theta$ of the true sieve coefficients, because the transformation $(\theta, \boldsymbol{p}) \to (\alpha\theta, \boldsymbol{p}^\alpha)$ preserves the FOC at every stake. The degenerate minimum at $\theta = 0$ corresponds to $\alpha \to 0$, which maps all beliefs to uniform. The cross-stake criterion alone cannot distinguish between different values of α; it identifies the direction of θ but not its magnitude.

Without any anchor ($\lambda_a = 0$), the estimator collapses to $\theta = 0$ in every simulation: mean L_1 error is 1.55, and 0% of agents improve over raw stated beliefs ($L_1 = 0.18$). The degenerate minimum is a global attractor.

D.3 Anchor weight sensitivity

The first key experiment varies the anchor penalty weight λ_a while keeping the anchor correctly centered (using each agent's true, noisily-observed RRA). This isolates the anchor's contribution as a function of its influence on the objective:

Anchor weight (λ_a)	Mean L_1	% agents improving
0.00 (no anchor)	1.55	0%
0.01	0.25	27%
0.10	0.11	79%
0.50	0.10	85%
1.00	0.10	85%
5.00	0.10	86%
20.0	0.10	86%
Stated baseline	0.18	—

The pattern is striking. Performance jumps discontinuously from catastrophic ($L_1 = 1.55$) to near-optimal ($L_1 = 0.11$) between $\lambda_a = 0.01$ and $\lambda_a = 0.10$, then plateaus. Beyond $\lambda_a = 0.10$, increasing the anchor weight by a factor of 200 (from 0.10 to 20.0) barely changes performance. This means the anchor's contribution is binary: it either breaks the degeneracy or it does not. Once the degeneracy is broken, the cross-stake variation carries the identifying information. The anchor weight beyond 0.10 adds virtually nothing.

D.4 Anchor centering sensitivity

The second experiment varies the *value* of the anchor, holding the weight fixed. This tests whether the anchor provides directional curvature information or merely serves as a regularizer:

Anchor scenario	Mean L_1	% agents improving
Correct individual anchor, $\lambda_a = 5$	0.10	86%
Correct individual anchor, $\lambda_a = 0.1$	0.11	79%
Wrong anchor (RRA = 1.0), $\lambda_a = 5$	0.21	39%
Wrong anchor (RRA = 1.0), $\lambda_a = 0.1$	0.23	30%
Wrong anchor (RRA = 3.0), $\lambda_a = 5$	0.46	11%
Wrong anchor (RRA = 3.0), $\lambda_a = 0.1$	0.43	13%
Generic anchor (RRA = 1.5), $\lambda_a = 0.1$	0.20	50%
Stated baseline	0.18	—

Anchor centering matters. A correctly centered anchor yields $L_1 \approx 0.10$ regardless of weight. A systematically wrong anchor (e.g., RRA = 3.0 for all agents) degrades performance well below the stated baseline, particularly at high weight. This rules out the interpretation that the anchor is "just a nudge": it provides genuine directional information about utility curvature, and that information must be approximately correct for the estimator to perform well.

A generic population-level anchor (RRA = 1.5, applied identically to all agents) avoids the catastrophic degeneracy and yields modest improvement over stated beliefs for about half the agents. It does not replicate the precision of a correctly centered individual anchor because the generic value is necessarily wrong for agents whose true RRA lies far from 1.5.

D.5 Interpretation

These results illustrate the power-scale ambiguity of Theorems 8 and 9 in action:

Theory. Cross-stake variation identifies the *direction* of the sieve coefficients θ (equivalently, the shape of ψ_i up to a positive scalar), but not the scale. This parallels the CRRA case, where $g_k^* \propto p_k^{1/\rho}$ requires knowing ρ to recover beliefs. The auxiliary risk measurement resolves the power-scale ambiguity by pinning $\alpha = 1$.

Estimation. The cross-stake criterion alone cannot distinguish between different values of α and is globally minimized at the degenerate $\alpha = 0$ endpoint ($\theta = 0$, uniform beliefs). The anchor breaks this degeneracy. Once α is approximately pinned, the overidentifying

Table 5: Ablation study: percent L_1 reduction and share improved, individual regime, by estimator variant. All variants use identical data, optimization, and routing; only the objective differs.

	Sieve Exact		Mild Misspec.		Strong Misspec.	
Estimator	% Red.	Impr.	% Red.	Impr.	% Red.	Impr.
SVD (null-space)	-11.1%	0.39	+1.5%	0.56	-9.7%	0.49
Anchor only ($\lambda_a > 0$, no CS)	+50.0%	0.95	+58.9%	0.85	+14.0%	0.63
CS only ($\lambda_a = 0$)	-114.8%	0.17	-336.9%	0.18	-358.3%	0.17
PMD full ($\lambda_a = 5$)	+51.5%	0.97	+69.7%	0.94	+44.0%	0.85

restrictions from multiple stakes provide the primary identifying signal for belief recovery, as evidenced by the anchor weight insensitivity beyond $\lambda_a = 0.10$.

Practical implication. The auxiliary risk measurement is theoretically necessary but need not be precise. Standard incentivized risk tasks (e.g., Holt–Laury) provide sufficient information, and the method degrades gracefully as anchor quality deteriorates. This robustness is relevant given that risk elicitation instruments are known to be noisy and sometimes context-dependent. When individual preference data are unavailable, a rough population-level anchor (e.g., from the literature or the experimental context) can serve as a fallback to prevent degeneracy, but it does not replicate the precision gains from a correctly centered individual measurement.

E Ablation Study

The PMD objective has two components: cross-stake consistency (CS) and the RRA anchor penalty. To determine each component's contribution, I decompose the estimator into four variants using identical data, optimization infrastructure, and CRRA routing. Only the objective function differs.

Three findings emerge. First, the cross-stake term alone (CS-only, $\lambda_a = 0$) is degenerate: without the anchor to pin scale, it converges to trivial solutions dramatically worse than the naive benchmark. Second, the anchor alone does most of the heavy lifting for easy DGPs (+50% and +59% for `sieve_exact` and `mild_misspec`). Third, the cross-stake term is essential for the hardest case: under `strong_misspec`, anchor-only achieves only +14% while PMD full achieves +44%. Both components contribute; neither is sufficient alone.

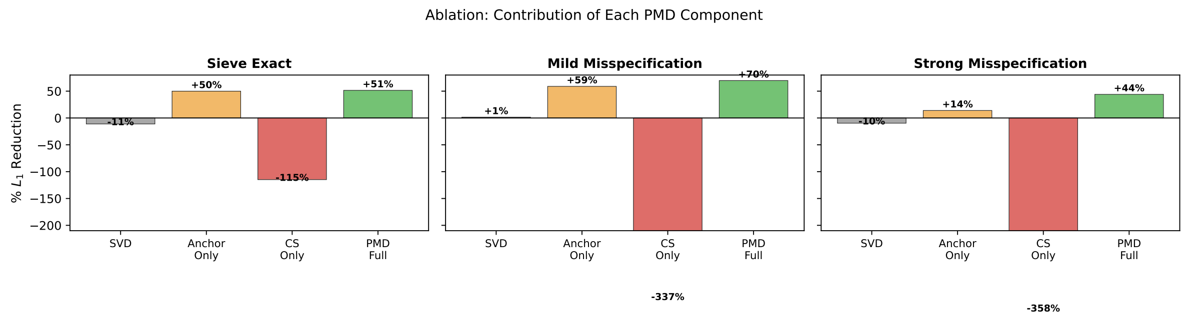

Figure 5: Ablation: percent L_1 reduction by estimator variant. CS-only collapses to degenerate solutions. The anchor does most of the work for easy DGPs, while the cross-stake term is essential under strong misspecification.

Table 6: Penalty weight sensitivity: λ_a from 0.1 to 50, individual regime, 500 agents × 50 replications.

	Mild Misspec.			Strong Misspec.		
λ_a	Rec. L_1	% Red.	Impr.	Rec. L_1	% Red.	Impr.
0.1	0.083	65.3%	0.93	0.092	43.8%	0.85
0.5	0.073	69.4%	0.94	0.092	43.8%	0.84
1.0	0.073	69.5%	0.94	0.093	43.4%	0.84
2.0	0.073	69.5%	0.94	0.093	43.2%	0.84
5.0	0.073	69.4%	0.94	0.092	43.9%	0.85
10	0.073	69.5%	0.94	0.092	44.3%	0.85
20	0.073	69.7%	0.94	0.092	44.4%	0.85
50	0.073	69.7%	0.94	0.092	44.4%	0.85

F Penalty Weight and Parameter Sensitivity

F.1 Penalty Weight

Performance is essentially flat: mild misspecification yields 65–70% reduction and strong misspecification yields 43–44% reduction across the entire λ_a range from 0.1 to 50.

F.2 Additional Sensitivity Analyses

Each analysis varies one parameter at a time from the baseline ($d = 2$, $M = 3$ stakes at $\{1, 3, 8\}$, anchor noise 0.08, $K = 10$, CRRA routing threshold 0.20), using `mild_misspec`.

Sieve dimension. Performance is similar for $d \in \{2, 3, 4\}$ (69.3%, 69.2%, and 68.7% respectively), confirming the low-dimensional sieve is sufficient.

Number of stakes. Two stakes yield 64% reduction; three reach 69%; five push to 73%.

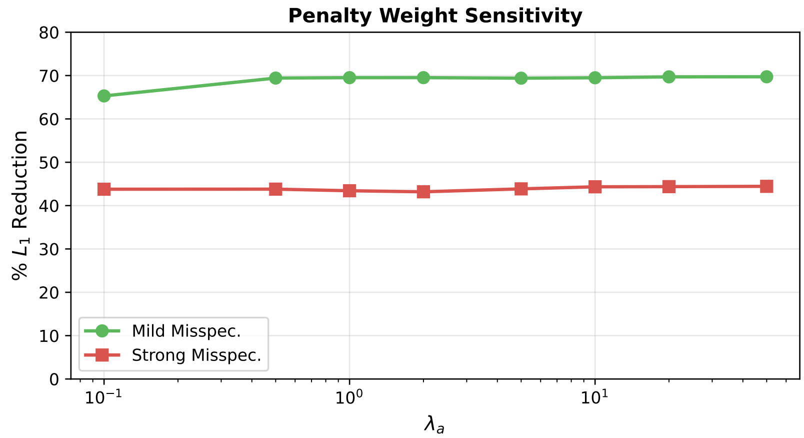

Figure 6: Percent L_1 reduction as a function of λ_a. Performance is essentially flat across a 500× range.

Table 7: Oracle-assisted versus feasible (non-oracle) CRRA routing in the robustness designs. "Feasible" uses the stake-invariance routing statistic only (no latent family labels). Here "positive" means assignment to the CRRA branch, so Feasible FP = share of non-CRRA agents routed to CRRA and Feasible FN = share of CRRA agents routed to the sieve branch. 500 agents × 50 replications.

DGP	Oracle % Red.	Feasible % Red.	Δ pp	Feasible FP	Feasible FN	Feasible Impr.
Sieve Exact	51.6%	51.5%	-0.2	0.92	0.09	0.97
Mild Misspec.	69.6%	68.7%	-1.0	0.66	0.09	0.93
Strong Misspec.	44.3%	41.4%	-2.9	0.44	0.09	0.82

Diminishing returns beyond $M = 3$ are consistent with theory.

Anchor noise. As noise increases (s.d. 0.04 to 0.60): 72%, 69%, 52%, and 8%. Even at three times baseline (s.d. 0.25), substantial improvement persists.

Number of bins. Recovery works across $K \in \{5, 10, 20\}$ (69%, 70%, 69%).

CRRA routing threshold. Stable across thresholds 0.12 to 0.28 (71% throughout). This indicates that aggregate recovery in this design is not highly sensitive to the routing cutoff over a wide range. Consistent with Remark 8, this should be interpreted as a tuning-robustness check, not as a formal power study of the stake-invariance classifier near the CRRA boundary.

Heaping noise. With 40% focal-point rounding to the nearest 0.05 grid: 44% (sieve exact), 53% (mild), 37% (strong)—modest degradation from normal-noise baseline.

Table 8: Near-CRRA routing exercise (weak-ID diagnostic): utilities drawn from a near-CRRA perturbation family with perturbation amplitude ϵ. Oracle and feasible routing achieve similar aggregate recovery, but the feasible classifier has high error rates near the CRRA boundary. "Positive" again denotes assignment to the CRRA branch (so FP and FN use the same convention as Table 7). 500 agents × 50 replications.

ϵ	Oracle % Red.	Feasible % Red.	Δ pp	Feasible FP	Feasible FN	Feasible Route Share	Feasible Impr.
0.02	58.7%	58.8%	+0.1	0.85	0.16	0.85	0.97
0.05	58.5%	58.8%	+0.3	0.84	0.16	0.84	0.97
0.10	58.7%	58.9%	+0.2	0.85	0.16	0.85	0.97
0.20	58.7%	58.9%	+0.1	0.84	0.16	0.84	0.97

F.3 Feasible Routing and Near-CRRA Weak-ID Diagnostics

Table 7 compares the oracle-assisted routing benchmark (used as an upper bound/diagnostic) with a feasible implementation that routes using only the stake-invariance statistic. Aggregate belief recovery is very similar: the feasible router loses 0.2 percentage points under `sieve_exact`, 1.0 point under `mild_misspec`, and 2.9 points under `strong_misspec`. At the same time, feasible routing produces substantial false positives, especially in easier DGPs. This pattern supports the practical recommendation in Remark 8: routing is a useful screening heuristic for computation, but not a classifier to interpret literally.

Table 8 provides a targeted weak-identification exercise around the CRRA boundary. Across perturbation amplitudes $\epsilon \in \{0.02, 0.05, 0.10, 0.20\}$, aggregate recovery is nearly unchanged under feasible routing relative to the oracle benchmark (differences of about 0.1–0.3 percentage points). However, the feasible routing rule exhibits high false-positive rates (about 0.84–0.85) and nontrivial false-negative rates (about 0.16), confirming that near-CRRA classification is weak in finite samples even when aggregate PMD performance is stable.

F.4 Basis Robustness (Log-Polynomial vs. Log-Spline)

Table 9 compares the baseline log-polynomial sieve with a log-spline basis in two flagship individual-recovery designs. Performance is nearly unchanged: the spline basis changes percent L_1 reduction by only +0.3 percentage points under `mild_misspec` and −0.4 points under `strong_misspec`. This targeted check does not exhaust basis-selection questions, but it indicates that the headline PMD gains are not an artifact of the log-polynomial specification.

Table 9: Targeted basis robustness check for flagship individual PMD designs: log-polynomial versus log-spline basis (both with $d = 3$). Appendix-scale run with 300 agents $\times$ 20 replications.

DGP	Basis	d	N	Rec. L_1	% Red.	Impr.
Mild Misspec.	Log-poly	3	300	0.074	68.7%	0.94
Mild Misspec.	Log-spline	3	300	0.074	69.0%	0.93
Mild Misspec.	Δ (spline - poly)	–	–	-0.000	+0.3	-0.00
Strong Misspec.	Log-poly	3	300	0.092	44.4%	0.85
Strong Misspec.	Log-spline	3	300	0.092	44.0%	0.84
Strong Misspec.	Δ (spline - poly)	–	–	+0.000	-0.4	-0.00

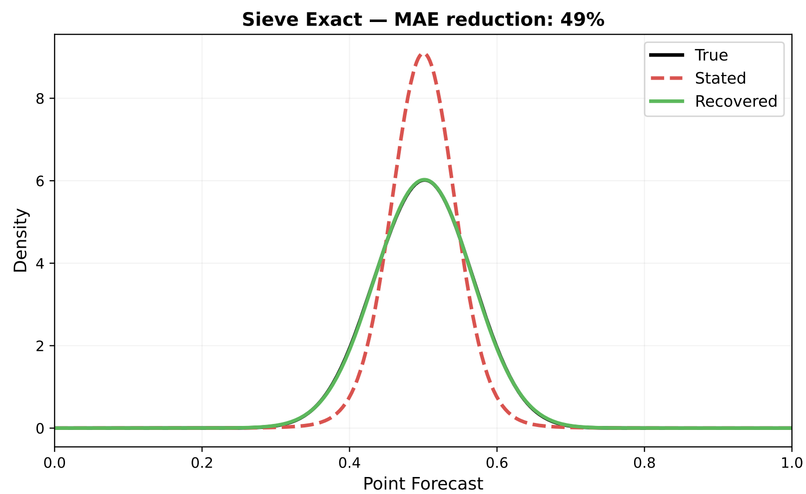

Figure 7: Distribution of point forecasts: `sieve_exact`. The recovered distribution (green) nearly perfectly overlaps the true distribution (black).

G Point Forecast Exercise

This section provides additional information about point forecast recovery. In particular, whenever reported expectations are biased by satisficing.

Under heaping noise, the stated distribution develops visible lumps at focal values, but the recovered distribution continues to approximate the true distribution.

H Convergence and Sieve Consistency

H.1 Finite-Sample Convergence

I run $N \in \{50, 100, 200, 500, 1000, 2000\}$ agents for each regime–DGP combination with 50 replications.

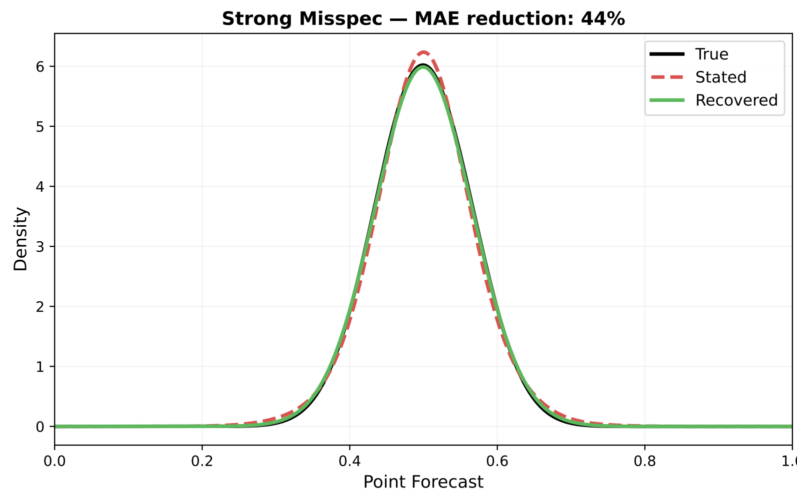

Figure 8: Distribution of point forecasts: `strong_misspec`. Recovery still substantially outperforms stated beliefs.

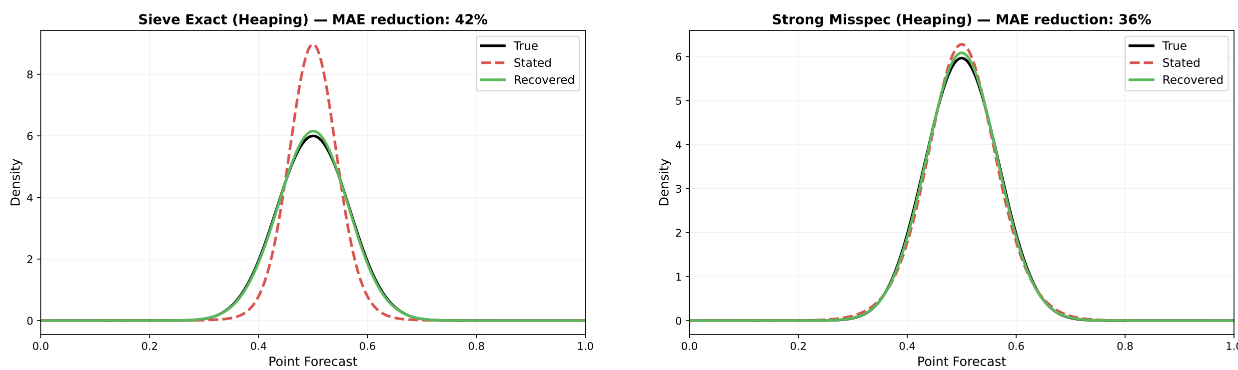

Figure 9: Point forecast distributions under heaping noise. Left: `sieve_exact`. Right: `strong_misspec`.

Three patterns emerge. First, the individual estimator is stable across all sample sizes (44% for strong misspecification at both $N = 50$ and $N = 2000$). Second, `common_wrong` drops from $+12\%$ at $N = 50$ to -86% at $N = 2000$ under strong misspecification—more data makes the misspecified pooled estimator converge to the wrong answer. Third, `common_correct` also degrades under strong misspecification (from $+31\%$ to -44%) due to finite-basis approximation bias: at large N, estimation noise vanishes, exposing the irreducible bias from projecting the true utility onto a degree-d sieve.

The degradation of `common_wrong` is a direct consequence of the identification structure in Theorem 9. Under heterogeneous utilities, each agent's power-scale parameter α_i (resolved by her individual anchor) differs. Pooling forces a single ψ and a single α on all agents. As N grows, the pooled estimator converges precisely to a compromise ψ that minimizes average

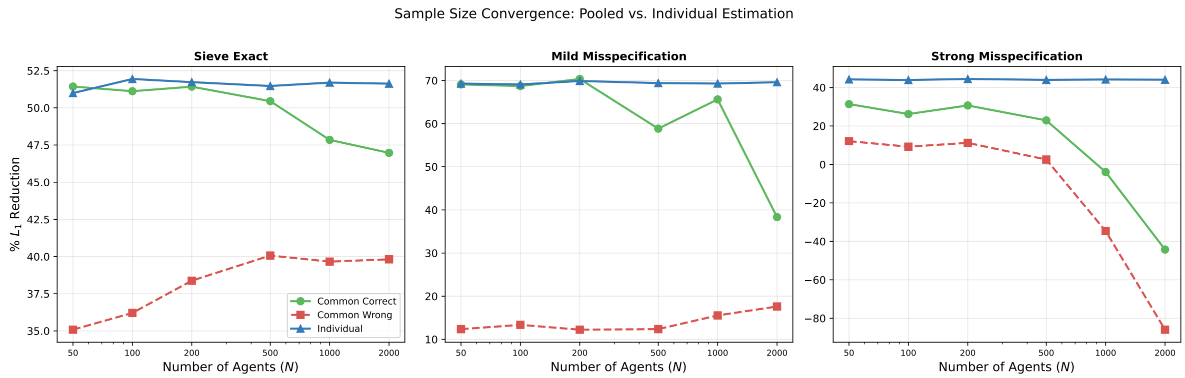

Figure 10: Convergence: percent L_1 reduction as a function of N, by regime and DGP. The individual estimator is stable across all sample sizes. Pooled estimators degrade at large N under misspecification.

loss across agents—a pseudo-true value that is correct for none of them. Each agent's beliefs are then systematically distorted: agents whose true α_i exceeds the pooled value have their beliefs sharpened, while those below have their beliefs flattened. More data makes this distortion more precise, not smaller. The individual estimator avoids this entirely because each agent's α is pinned by her own anchor independently.

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